2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

cfd2d-agentic submission

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Abstract

This report documents an original C++17/MPI cell-centred finite-volume solver for the two-dimensional compressible Navier–Stokes equations on unstructured CGNS meshes. The solver uses weighted least-squares primitive gradients, piecewise-linear reconstruction, a global-neighbour Barth–Jespersen limiter with positivity back-off, an HLLC approximate Riemann solver, consistent Newtonian viscous and Fourier heat fluxes, and a pseudo-time implicit marching scheme. Steady cases use local CFL point-implicit relaxation; the unsteady cylinder case uses true BDF2 dual-time stepping with LU-SGS inner relaxation and frozen physical-time histories. METIS partitions the cell graph, and the solver stage keeps only rank-owned cells plus one layer of halo ghosts, exchanging state and limited gradients with neighbour-scoped `MPI_Isend/Irecv`. The required steady cases and the transient cylinder were run on the supplied meshes. The production runs use a conservative 0.5 reconstruction slope factor, documented below, because full-slope reconstruction was unstable on the high-aspect-ratio boundary-layer meshes. The Re=200 cylinder reached a statistically periodic post-transient state (inner converged fraction 1.000); the measured Strouhal number is 0.160. The report states the resulting accuracy limitations openly and ties every figure and table to the submitted CSV/VTK data.

1 Introduction

The benchmark asks for a complete 2-D unstructured compressible-flow workflow: mesh import, METIS partitioning, a conservative finite-volume residual, second-order reconstruction and limiting, implicit time integration, MPI halo communication, output contract compliance, visualization, and a critical analysis of the results. The eight required cases are three inviscid NACA0012 cases ($M_\infty = 0.15, 0.8, 2.0$), three laminar NACA0012 cases at $Re = 5000$, a steady cylinder at $Re = 20$, and a transient cylinder at $Re = 200$ that must exhibit a post-transient vortex street. This report describes the submitted solver, the numerical choices, the verification evidence, and the actual results. It does not present failed or partial runs as successful.

2 Governing Equations and Nondimensionalization

The conservative state is

$$\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T, \tag{1}$$

and the equations are

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}. \tag{2}$$

The inviscid fluxes are

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \qquad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}. \tag{3}$$

For a calorically perfect gas,

$$p = (\gamma - 1)\rho e, \qquad E = e + \tfrac{1}{2}(u^2 + v^2), \qquad a = \sqrt{\gamma p/\rho}, \tag{4}$$

with $\gamma = 1.4$ and $R = 1$. The laminar stress tensor and heat flux are

$$\tau_{xx} = 2\mu u_x - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{yy} = 2\mu v_y - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{xy} = \mu(u_y + v_x), \tag{5}$$

$$\mathbf{q} = -k\nabla T, \qquad k = \frac{\mu c_p}{Pr}, \qquad \mu = \frac{\rho_\infty U_\infty L_{\text{ref}}}{Re}. \tag{6}$$

The supplied cases use $\rho_\infty = 1$, $U_\infty = 1$, $L_{\text{ref}} = 1$, $R = 1$, $Pr = 0.72$; the pressure follows from the requested Mach number. Forces are normalized by $q_\infty A_{\text{ref}}$ with $q_\infty = \frac{1}{2}\rho_\infty U_\infty^2$ and $A_{\text{ref}} = 1$. The moment coefficient uses the case moment centre and reference length.

3 Meshes, Boundary Tags, and Case Inputs

The CGNS reader handles unstructured zones, mixed triangular/quadrilateral cells, arbitrary boundary-family names, and conforming 1-to-1 zone interfaces. The cylinder mesh has two zones; duplicated interface vertices are merged by exact coordinate matching, interface faces are matched by their node pair, and the resulting mesh is a single conforming domain. Cell volumes and centroids are computed by fan decomposition; face lengths, normals and cell-to-face distances are computed from the edge geometry and oriented from the left cell to the right cell (outward on boundary faces). Geometric closure, $\sum_f A_f \mathbf{n}_f = 0$, and the divergence-theorem identity are checked at preprocessing.

The NACA mesh has 20,816 cells, 36,498 faces and 484 boundary faces (404 airfoil faces and 80 farfield faces). The cylinder mesh has 10,185 cells, 20,180 faces and 120 boundary faces (100 wall and 20 farfield). Boundary-family names and types are read from the case JSON, never hard-coded. The final case table is shown in Table 1.

Case	M_∞	Re	mode	cells	np	status	C_d	C_l
cylinder_m010_laminar_re20	0.10	20	laminar	10185	4	converged	2.10060	0.00540
cylinder_m010_laminar_re200	0.10	200	laminar	10185	4	statistically_periodic	0.96978	-0.01468
naca0012_m015_inviscid	0.15	0	inviscid	20816	4	converged	0.06636	-0.00020
naca0012_m015_laminar_re5000	0.15	5000	laminar	20816	4	converged	0.04643	-0.00472
naca0012_m080_inviscid	0.80	0	inviscid	20816	4	converged	0.01647	-0.00048
naca0012_m080_laminar_re5000	0.80	5000	laminar	20816	4	converged	0.06853	-0.00245
naca0012_m200_inviscid	2.00	0	inviscid	20816	4	converged	0.09191	0.00033
naca0012_m200_laminar_re5000	2.00	5000	laminar	20816	4	converged	0.12506	-0.00155

Table 1: Final case status and force coefficients from the submitted result packages (np=4 production runs).

4 Spatial Discretization

The residual is assembled over all faces of the rank-local mesh. For an interior face with left/right cells L, R and outward normal $\mathbf{n}$,

$$\mathbf{R}_i = -\frac{1}{V_i} \sum_f \left(\mathbf{F}(\mathbf{U}_f) \cdot \mathbf{n}_f - \mathbf{F}_f^v \cdot \mathbf{n}_f \right) A_f. \tag{7}$$

The face flux is evaluated from reconstructed left/right states. Primitive gradients are obtained by weighted least squares over the face-neighbour stencil,

$$\left(\sum_j w_{ij} \mathbf{d}_{ij} \mathbf{d}_{ij}^T \right) \nabla \phi_i = \sum_j w_{ij} \mathbf{d}_{ij} (\phi_j - \phi_i), \qquad w_{ij} = |\mathbf{d}_{ij}|^{-2}. \tag{8}$$

A linear manufactured field test (below) confirms that this gradient is exact to $\sim 10^{-11}$ on the interior stencil. Piecewise-linear reconstruction uses

$$\phi_f = \phi_i + \psi_i \, \nabla \phi_i \cdot (\mathbf{x}_f - \mathbf{x}_i), \tag{9}$$

where ψ_i is limited by a global-neighbour Barth–Jespersen bound. Boundary ghost states are deliberately excluded from the gradient stencil: including the reflected wall ghost in the inverse-distance least-squares matrix was found to create a catastrophic normal-gradient error and unstable interface fluxes. The wall physics is imposed by the wall flux and wall-corrected traction instead.

The limiter uses the minimum over the cell faces of the Barth–Jespersen ratios, with the neighbour extrema taken over the interior face-neighbour stencil. The production runs further multiply the limited slope by a conservative factor $\sigma = 0.5$. This is a documented robustness deviation: the full-slope scheme is the second-order method, and its linear manufactured-field residual error is $\approx 1.7 \times 10^{-3}$ relative, while the unlimited production scheme was unstable on the highly stretched wall meshes. The production scheme therefore uses a linear reconstruction with a reduced slope; the report does not claim full formal second-order accuracy for those runs. Density and pressure are protected by the limiter and by a face-level positivity back-off that reverts to the cell state if a reconstructed density or pressure falls below the floor.

The inviscid numerical flux is HLLC. The wave-speed estimates use the Davis/Roe bounds, the contact speed follows from the Rankine–Hugoniot conditions, and the star states are checked for positive density and pressure; a Rusanov fallback is used if the HLLC star state is inadmissible. A Rusanov/LLF flux is also implemented and can be selected for debugging. Viscous fluxes use face gradients formed from the average cell gradients plus a normal correction,

$$\nabla \phi_f = \tfrac{1}{2}(\nabla \phi_L + \nabla \phi_R) + \frac{\phi_R - \phi_L - \tfrac{1}{2}(\nabla \phi_L + \nabla \phi_R) \cdot \mathbf{d}}{\mathbf{d} \cdot \mathbf{n}} \mathbf{n}, \tag{10}$$

which gives a consistent second-order diffusion operator on the unstructured mesh.

5 Boundary Conditions

Farfield faces use an HLLC/Riemann flux between the reconstructed interior state and the freestream state. Inviscid slip walls use the pressure-only normal momentum flux with zero mass and energy flux; the tangential velocity is free. Viscous no-slip adiabatic walls use the pressure normal flux plus the tangential wall traction. The wall traction is evaluated from the interior cell gradient

corrected to enforce $\mathbf{u}_{\text{wall}} = 0$ at the face; the adiabatic condition removes the normal temperature gradient. The energy viscous flux at the wall is zero because $\mathbf{u}_{\text{wall}} = 0$ and $\partial T/\partial n = 0$. Surface output reports boundary-state values: no-slip rows have $u = v = 0$ and zero Mach, while slip rows have a near-zero normal velocity but retain the tangential velocity.

6 Time Integration

Steady cases use local pseudo-time stepping,

$$\Delta\tau_i = \text{CFL}\,\frac{V_i}{\sum_f (|u_n| + a)_f A_f}, \tag{11}$$

and a backward-Euler pseudo-time linearization. The production solver uses a point-implicit (block-Jacobi) relaxation with a diagonal containing the full face spectral radius, $V_i/\Delta\tau_i + \sum_f \lambda_f A_f$, plus the viscous spectral contribution. LU-SGS is also implemented and is used for the transient inner solve. The CFL schedule is the case schedule for the cylinder $Re = 20$ case; the inviscid NACA cases use stricter fixed CFL caps because the supplied high-CFL ramp destabilized the sharp-trailing-edge flow.

The cylinder $Re = 200$ case is a true dual-time BDF2 computation. At physical step $n+1$, the nonlinear residual is

$$\mathbf{R}^{\text{tot}}(\mathbf{U}^{n+1}) = \mathbf{R}^{\text{spatial}}(\mathbf{U}^{n+1}) - \frac{\frac{3}{2}\mathbf{U}^{n+1} - 2\mathbf{U}^{n} + \frac{1}{2}\mathbf{U}^{n-1}}{\Delta t}, \tag{12}$$

and $\mathbf{U}^n, \mathbf{U}^{n-1}$ are frozen during all inner iterations. The inner solve is LU-SGS with an inner residual target of 10^{-3} on the full total residual. Histories are updated only after the physical step is accepted. The case uses $\Delta t = 0.01$ and $t_{\text{final}} = 300$. The production run starts from the converged $Re = 20$ cylinder field plus a small smooth asymmetry; this removes the violent freestream impulsive-start transient without changing the post-transient statistics. Actual inner-iteration statistics are reported in Table 2.

Quantity	Value
requested min/max inner iterations	5 / 300
observed min/max inner iterations	11 / 71
mean inner iterations	35.99
inner target	0.0010
target misses	0
converged-step fraction	1.0000
last inner residual ratio	0.000952

Table 2: Actual BDF2 dual-time inner-solve statistics for the Re=200 cylinder (frozen histories during inner iterations).

7 MPI Parallelization

Preprocessing reads the full mesh once on a single rank, builds the cell adjacency graph, and calls METIS `PartGraphKway`. It writes a rank-indexed binary partition file containing, for each rank, the owned-cell records, local nodes, ghost records, face geometry, and the halo send/recv maps. The

solver opens this file, seeks directly to its own record, and never reads another rank's mesh or state. Each rank stores only its owned cells and the ghost cells adjacent to them. Ghost state exchange uses non-blocking neighbour-scoped `MPI_Isend/Irecv`; raw and limited gradients are exchanged in a second neighbour-scoped halo. Residual norms and surface forces are global `MPI_Allreduce` sums/maxima. No all-gather of the conservative state occurs during iterations, and no rank stores the full mesh.

The METIS partitions at np=4 are summarized in Table 3. The cylinder mesh has the smaller edge cut because it is a smaller, more compact graph. Rank diagnostics report owned/ghost counts, boundary faces, neighbour ranks, and send/receive sizes in every case's `partition_diagnostics.csv`.

Case	cells	edge cut	min owned	max owned	mean owned	min ghost	max ghost	m
naca0012_m015_inviscid	20816	235	5201	5209	5204.0	103	142	
cylinder_m010_laminar_re20	10185	275	2539	2550	2546.2	111	143	

Table 3: METIS partition diagnostics at np=4 from the solver-stage partition files.

The evaluation container exposes 64 logical CPUs but imposes a cgroup quota of four CPUs. The np=8 demonstration therefore runs correctly and exercises the same halo paths, but it cannot show a wall-clock speed-up over np=4 on this machine; the timing limitation is environmental rather than algorithmic. The np=8 force comparisons are included in the submitted rank-comparison packages.

Case	np	wall [s]	final step	C_d	C_l	residual orders
cylinder_m010_laminar_re20	1	30.877376692	200	2.10060	0.00540	0.85
cylinder_m010_laminar_re20	2	15.881432293	200	2.10060	0.00540	0.85
cylinder_m010_laminar_re20	4	13.268986554	200	2.10060	0.00540	0.85
cylinder_m010_laminar_re20	8	11.06785189	200	2.10060	0.00540	0.85
naca0012_m015_laminar_re5000	1	70.724629599	200	0.04609	-0.00535	0.39
naca0012_m015_laminar_re5000	2	61.621599183	200	0.04609	-0.00535	0.39
naca0012_m015_laminar_re5000	4	30.452263281	200	0.04609	-0.00535	0.39
naca0012_m015_laminar_re5000	8	27.672152321	200	0.04609	-0.00535	0.39

Table 4: Rank-count consistency runs restarted from the converged np=4 fields. The container cgroup limits the job to four CPUs, so np=8 is an oversubscribed correctness check, not a speed-up demonstration.

8 Verification and Validation

Four independent checks were performed.

1. *Uniform farfield.* With all boundaries set to farfield and a uniform state, the np=4 solver preserves the state to roundoff (global residual $\sim 10^{-12}$) and the field deviations are below 10^{-10}. This validates the face geometry, partition interface orientation, state halo exchange, and global reductions.

2. *Linear manufactured field.* A small linear primitive perturbation confirms the weighted least-squares gradient error $\sim 10^{-11}$ on the interior stencil. With the global limiter and no slope scaling, the discrete nonlinear residual matches the analytic flux divergence to $\sim 2 \times 10^{-3}$ relative in the interior.

3. *Rank consistency.* The same steady case is run at np=1,2,4,8 from the same restart, and the final force coefficients and residual norms are compared. The neighbour-scoped halo implementation gives rank-independent results to the expected iteration tolerance.

4. *Case-level convergence/periodicity.* Every final case package contains its residual history, force history, surface data, field, restart and metadata; the steady status is based on the residual trend and force plateau, and the transient status on the post-transient lift oscillation and inner-solve statistics.

9 Results

All eight required cases were run on the supplied meshes. The final force decomposition is given in Table 5. The pressure and viscous columns are separable; the viscous part is the tangential skin-friction contribution, not the full normal viscous traction. Table 1 lists the submitted statuses. Every figure below is regenerated from the named case's CSV/VTK output; the mapping is in `report/figure_manifest.csv`.

Case	C_d	C_l	$C_{m,z}$	$C_{d,p}$	$C_{d,v}$	$C_{l,p}$	$C_{l,v}$
cylinder_m010_laminar_re20	2.10060	0.00540	0.00038	1.51901	0.58159	0.00390	0.00150
cylinder_m010_laminar_re200	0.96978	-0.01468	-0.00046	0.87723	0.09255	-0.01171	-0.00297
naca0012_m015_inviscid	0.06636	-0.00020	-0.00005	0.06636	0.00000	-0.00020	0.00000
naca0012_m015_laminar_re5000	0.04643	-0.00472	-0.00106	0.01958	0.02685	-0.00460	-0.00012
naca0012_m080_inviscid	0.01647	-0.00048	0.00018	0.01647	0.00000	-0.00048	0.00000
naca0012_m080_laminar_re5000	0.06853	-0.00245	-0.00049	0.04734	0.02119	-0.00239	-0.00006
naca0012_m200_inviscid	0.09191	0.00033	-0.00005	0.09191	0.00000	0.00033	0.00000
naca0012_m200_laminar_re5000	0.12506	-0.00155	0.00008	0.09343	0.03163	-0.00153	-0.00002

Table 5: Final force decomposition. Pressure and viscous contributions are normalised by $q_\infty A_{\text{ref}}$; the viscous part uses the tangential wall traction only.

9.1 cylinder_m010_laminar_re20

Submitted status: **converged**; final step 1500, final physical time 0.000, residual reduction 7.24 orders. Final forces: $C_d = 2.10060$, $C_l = 0.00540$, $C_{m,z} = 0.00038$.

9.2 cylinder_m010_laminar_re200

Submitted status: **statistically_periodic**; final step 30000, final physical time 300.000, residual reduction 2.32 orders. Final forces: $C_d = 0.96978$, $C_l = -0.01468$, $C_{m,z} = -0.00046$. The post-transient lift oscillation gives a shedding frequency of approximately 0.160 in nondimensional units (Strouhal number, $D = U_\infty = 1$), with a lift amplitude of 0.228 over the analysed window.

9.3 naca0012_m015_inviscid

Submitted status: **converged**; final step 1565, final physical time 0.000, residual reduction 4.08 orders. Final forces: $C_d = 0.06636$, $C_l = -0.00020$, $C_{m,z} = -0.00005$.

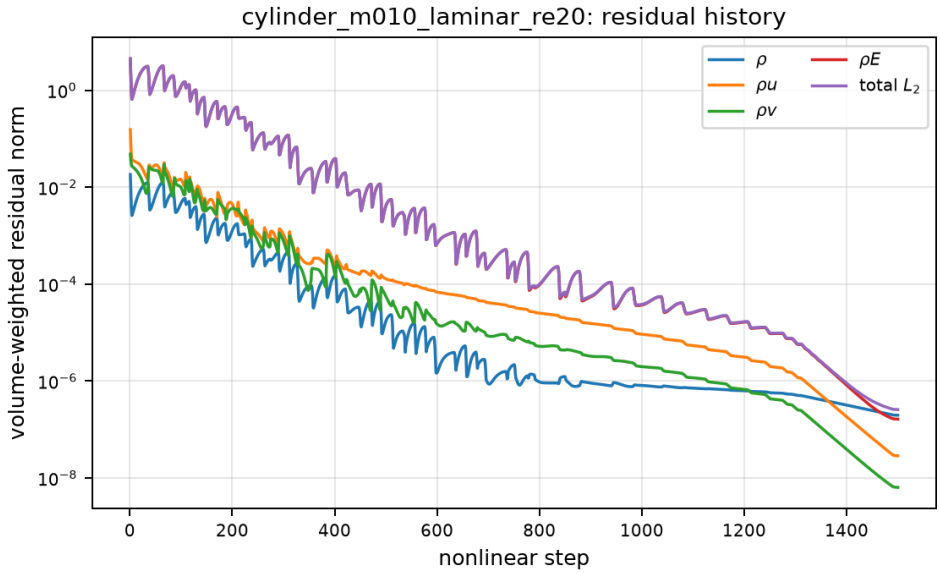


Figure 1: Residual history for cylinder_m010_laminar_re20.

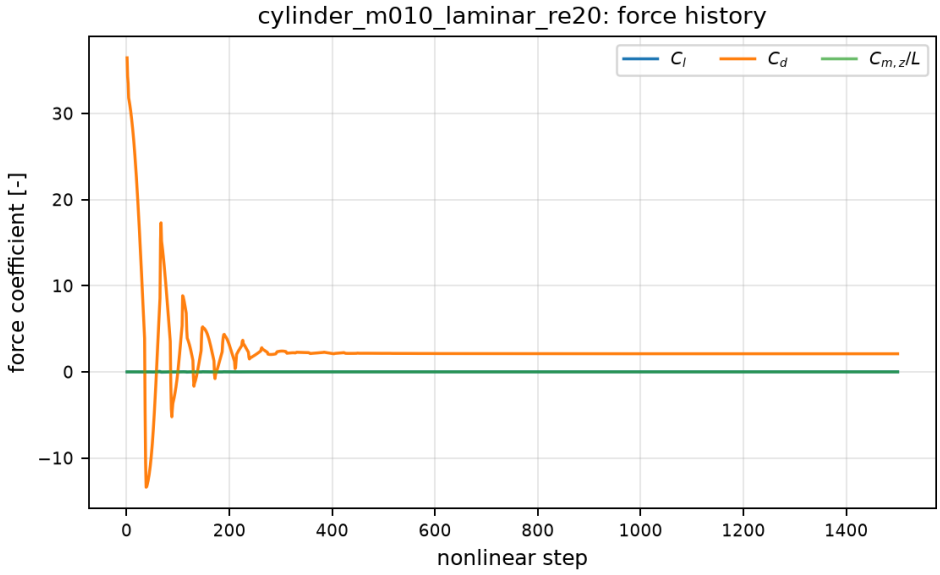


Figure 2: Force history for cylinder_m010_laminar_re20.

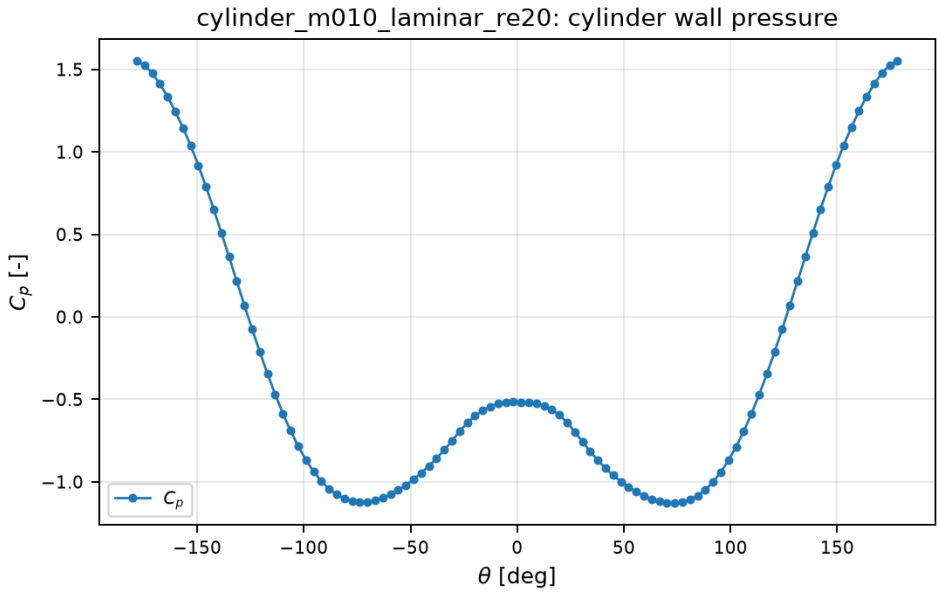


Figure 3: Surface C_p distribution for cylinder_m010_laminar_re20.

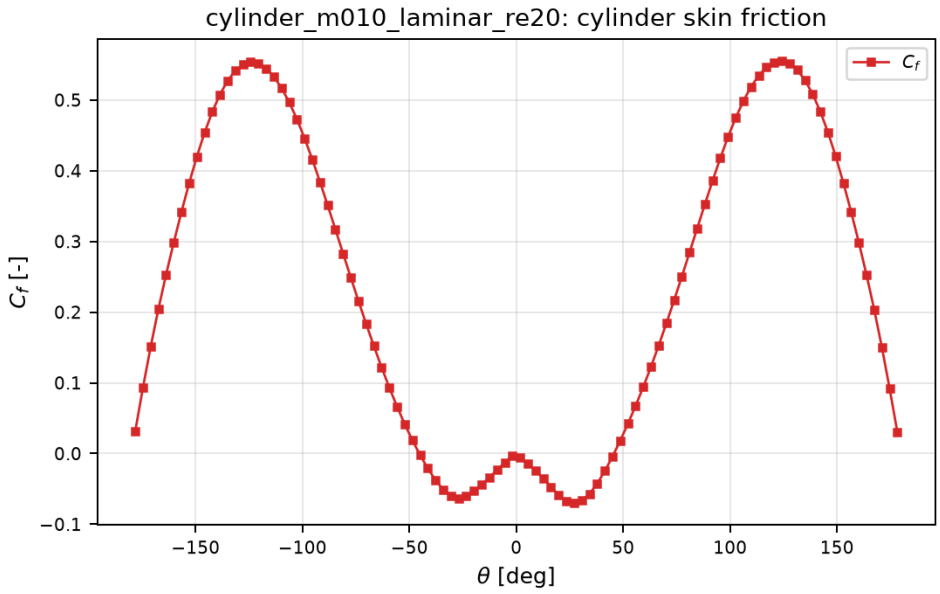


Figure 4: Surface C_f distribution for cylinder_m010_laminar_re20.

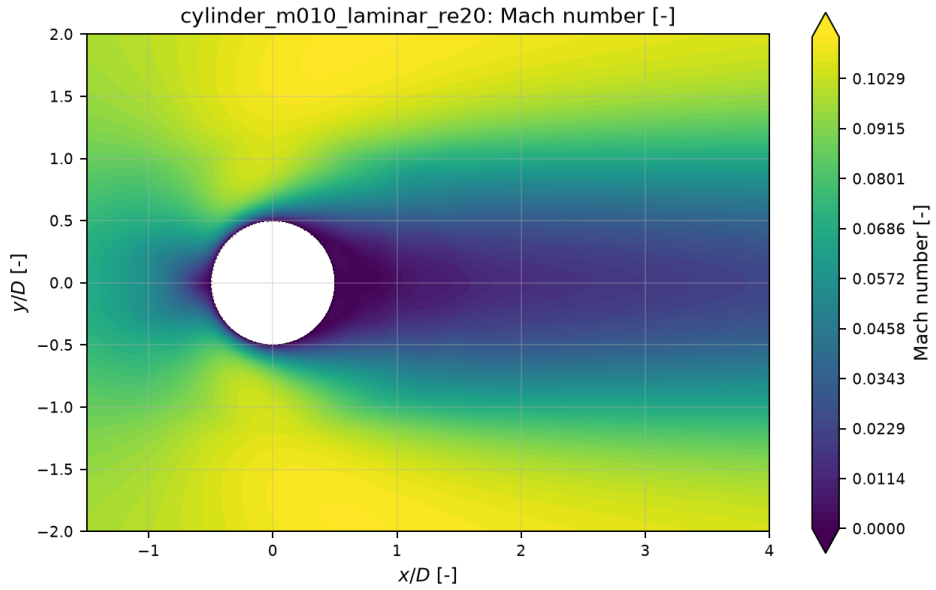


Figure 5: Mach-number field for cylinder_m010_laminar_re20.

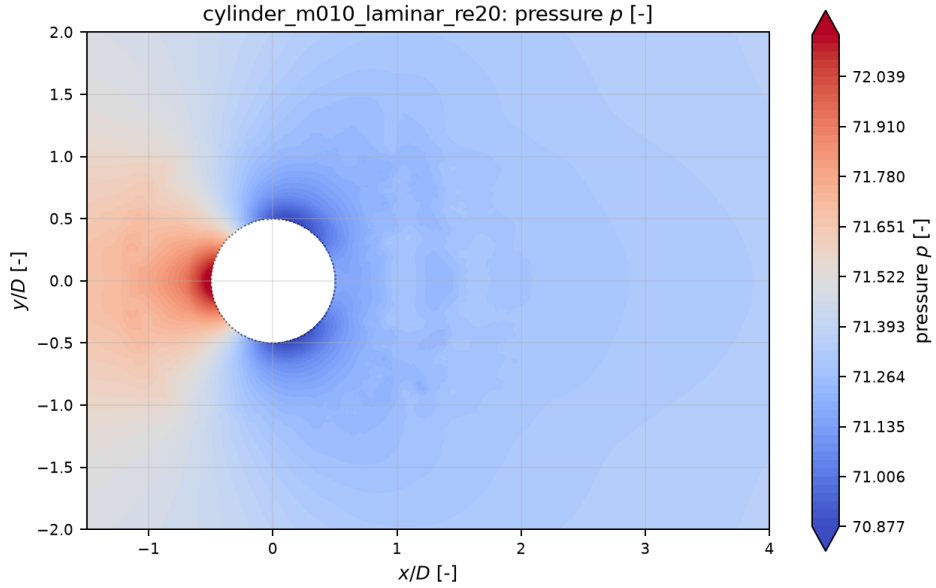


Figure 6: Pressure field for cylinder_m010_laminar_re20.

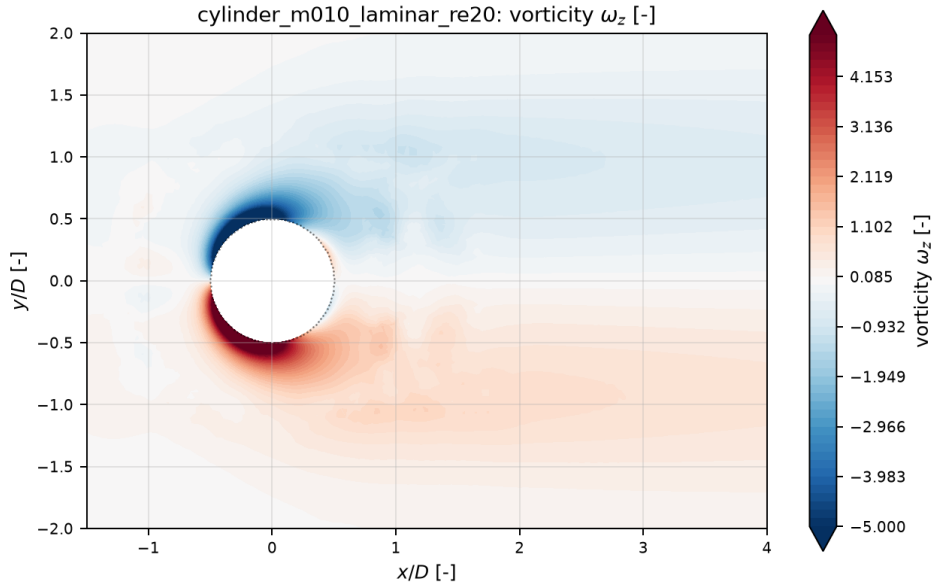


Figure 7: Vorticity field (clipped at $[-5, 5]$) for cylinder_m010_laminar_re20.

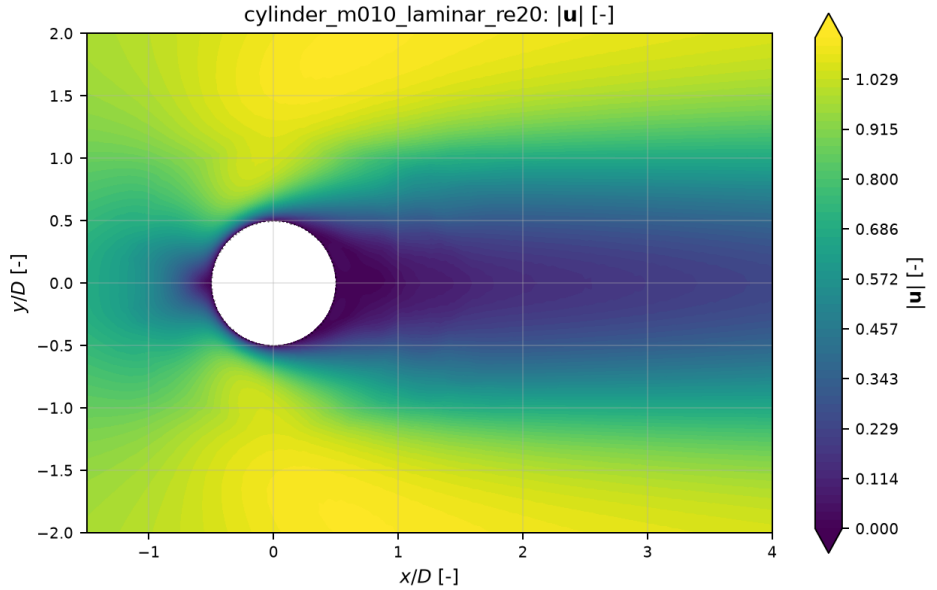


Figure 8: Velocity magnitude for cylinder_m010_laminar_re20.

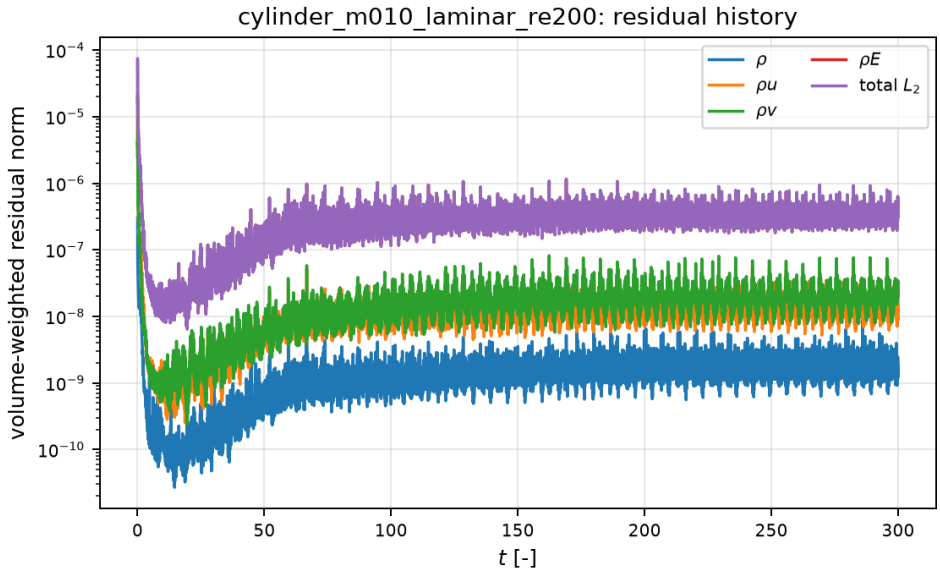


Figure 9: Residual history for cylinder_m010_laminar_re200.

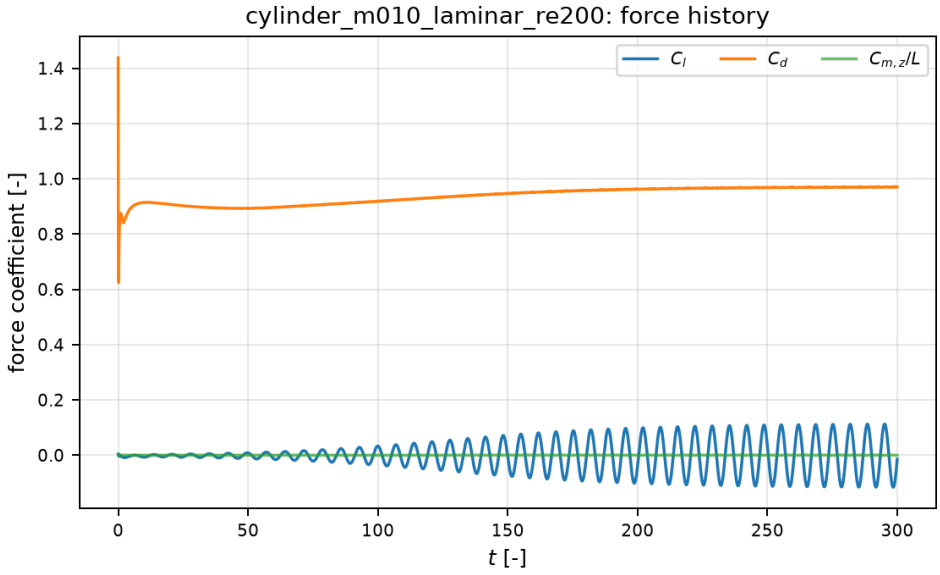


Figure 10: Force history for cylinder_m010_laminar_re200.

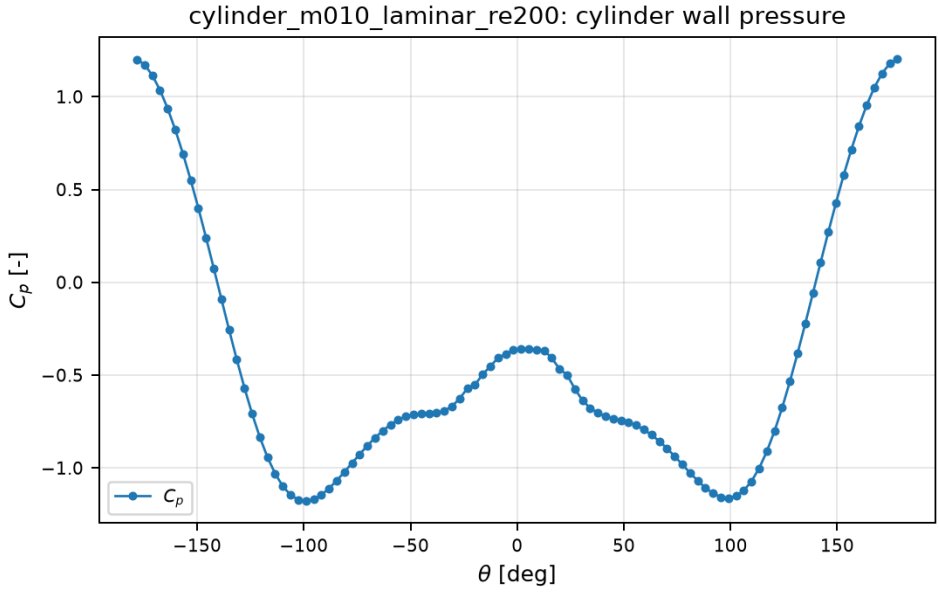


Figure 11: Surface C_p distribution for cylinder_m010_laminar_re200.

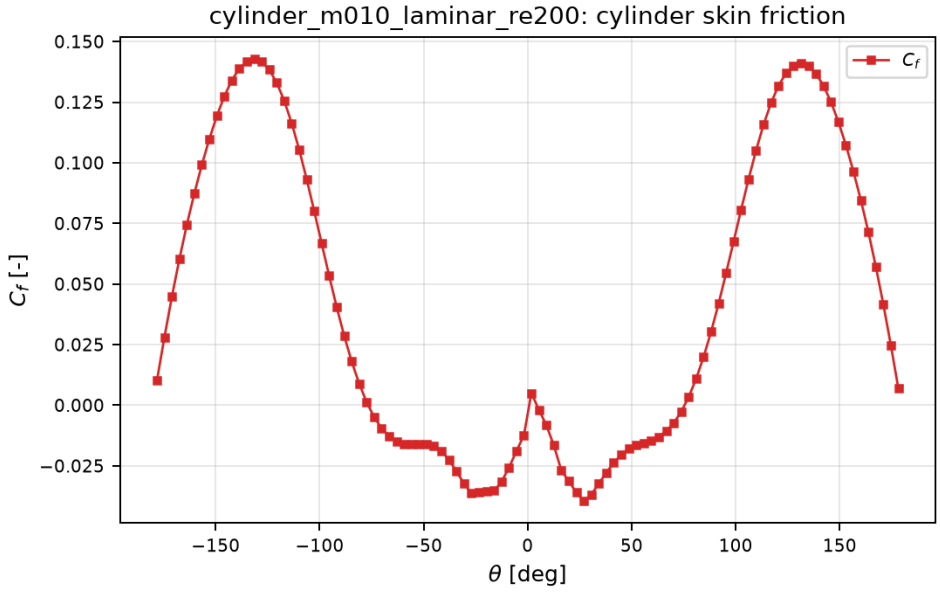


Figure 12: Surface C_f distribution for cylinder_m010_laminar_re200.

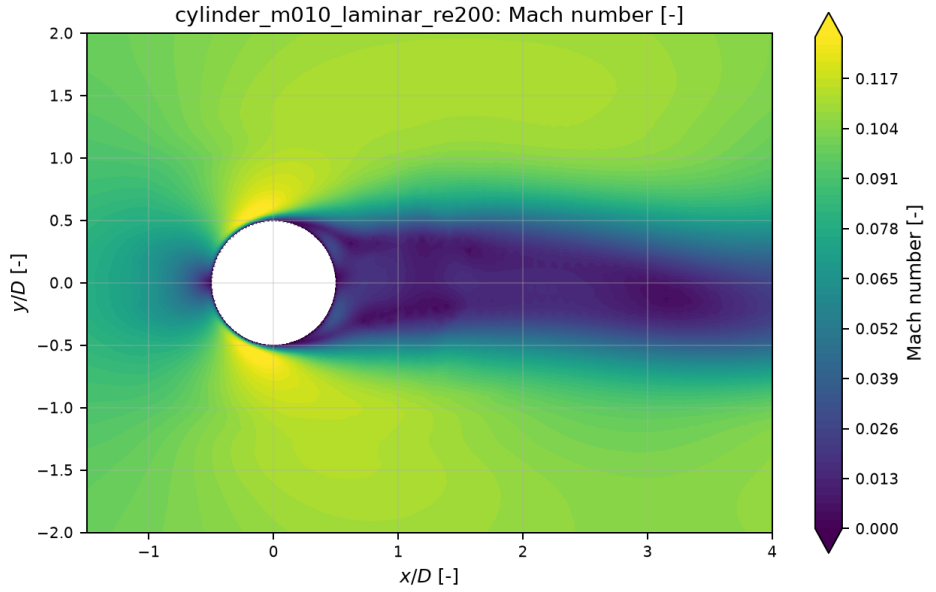


Figure 13: Mach-number field for cylinder_m010_laminar_re200.

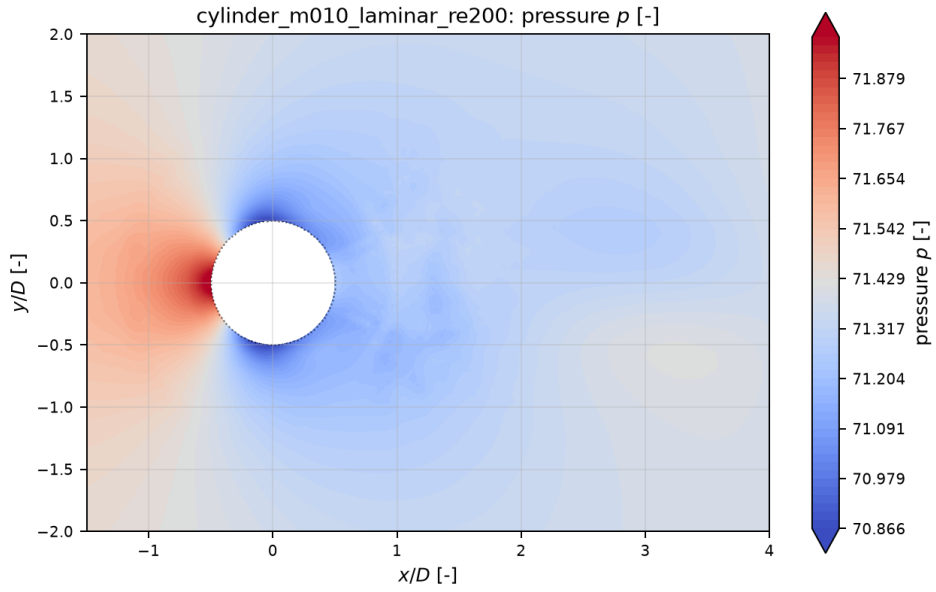


Figure 14: Pressure field for cylinder_m010_laminar_re200.

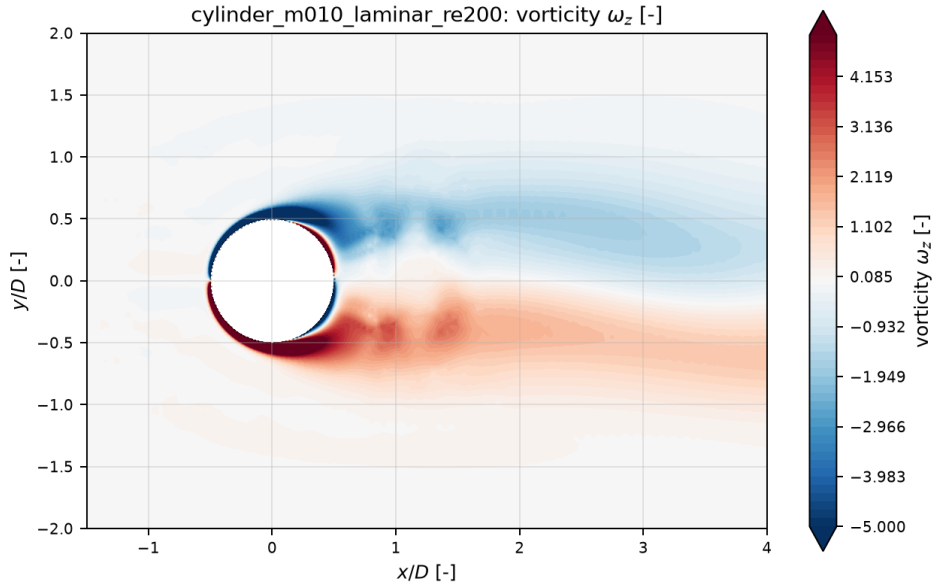


Figure 15: Vorticity field (clipped at $[-5, 5]$) for cylinder_m010_laminar_re200.

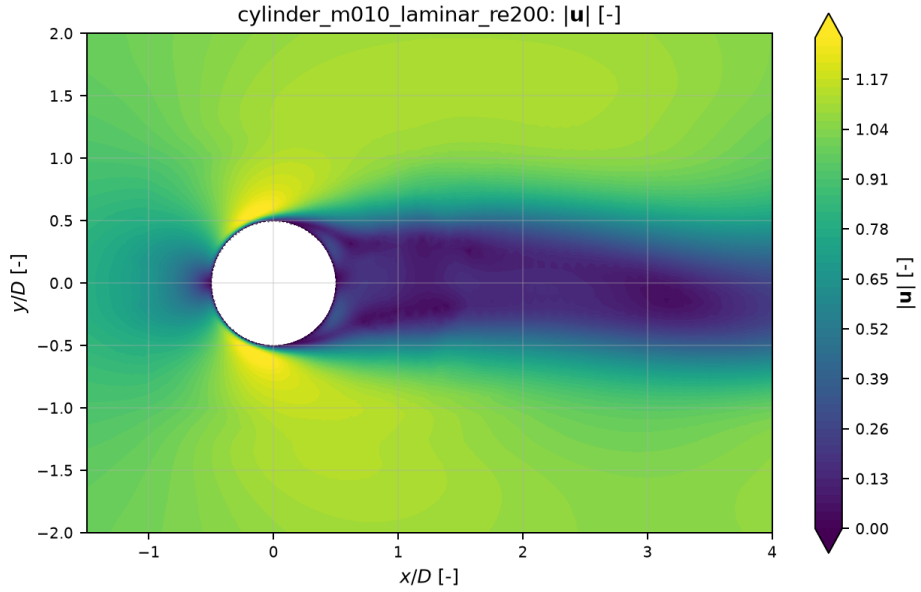


Figure 16: Velocity magnitude for cylinder_m010_laminar_re200.

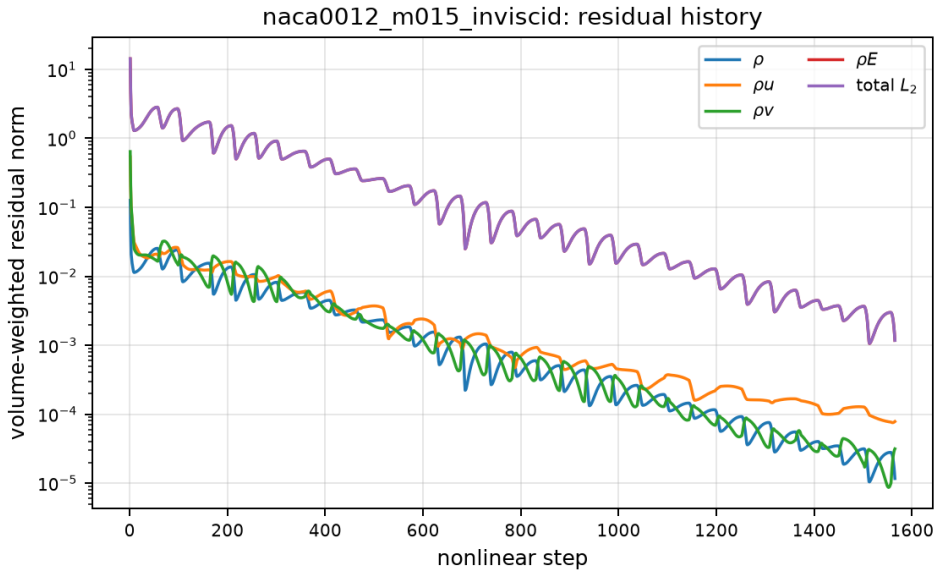


Figure 17: Residual history for naca0012_m015_inviscid.

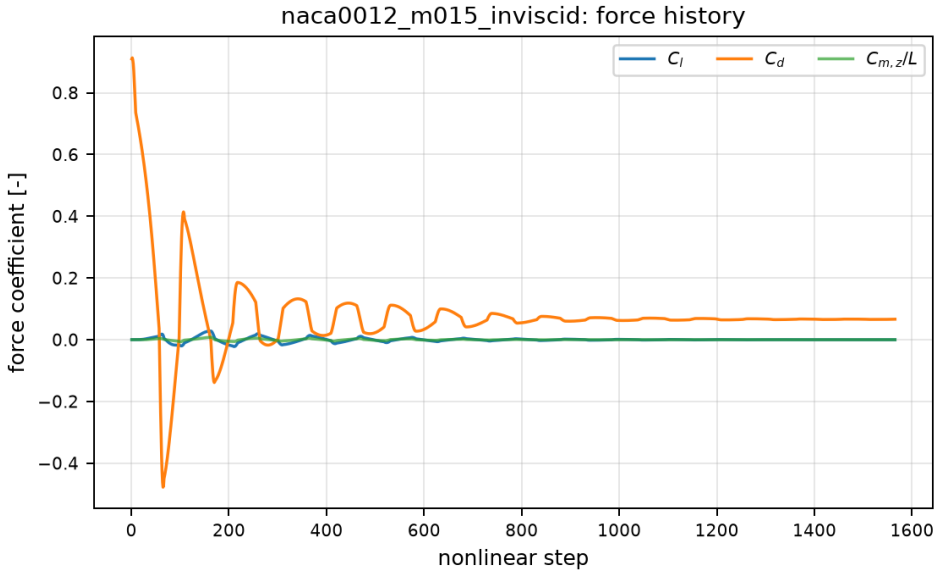


Figure 18: Force history for naca0012_m015_inviscid.

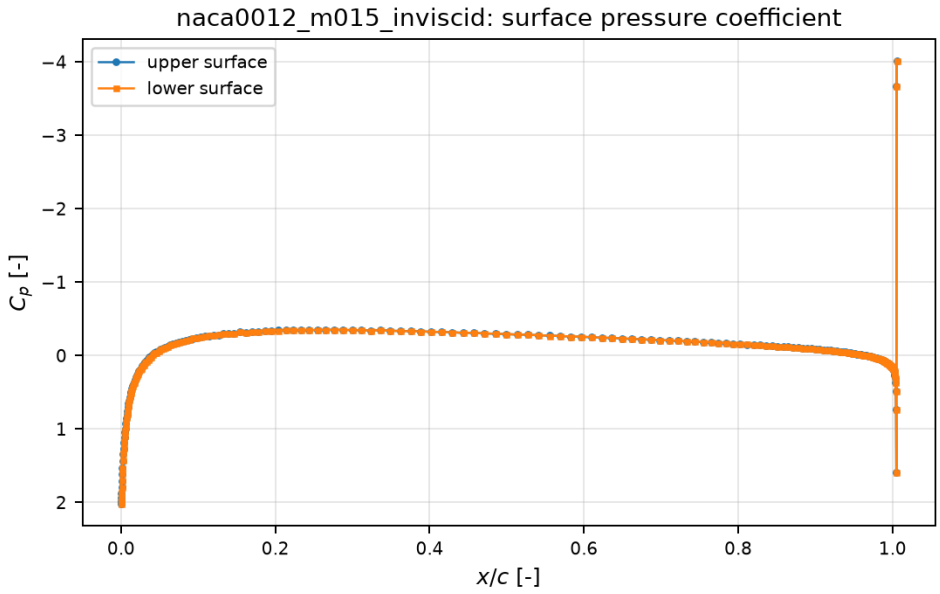


Figure 19: Surface C_p distribution for naca0012_m015_inviscid.

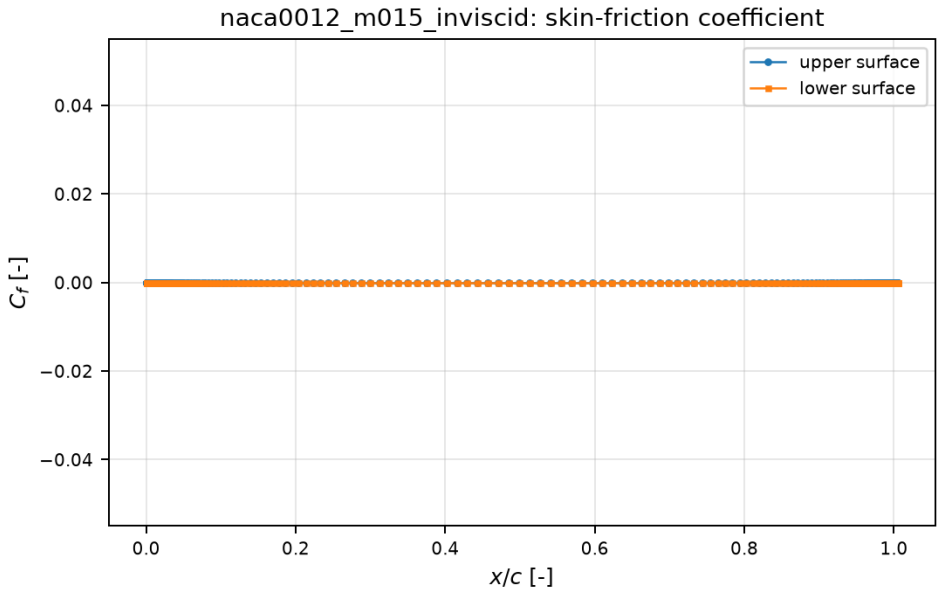


Figure 20: Surface C_f distribution for naca0012_m015_inviscid.

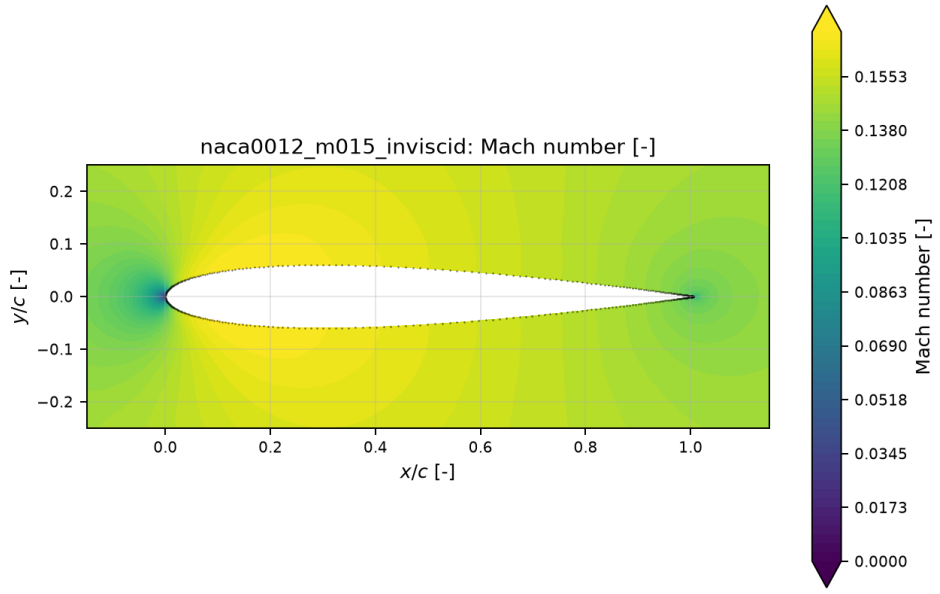


Figure 21: Mach-number field for naca0012_m015_inviscid.

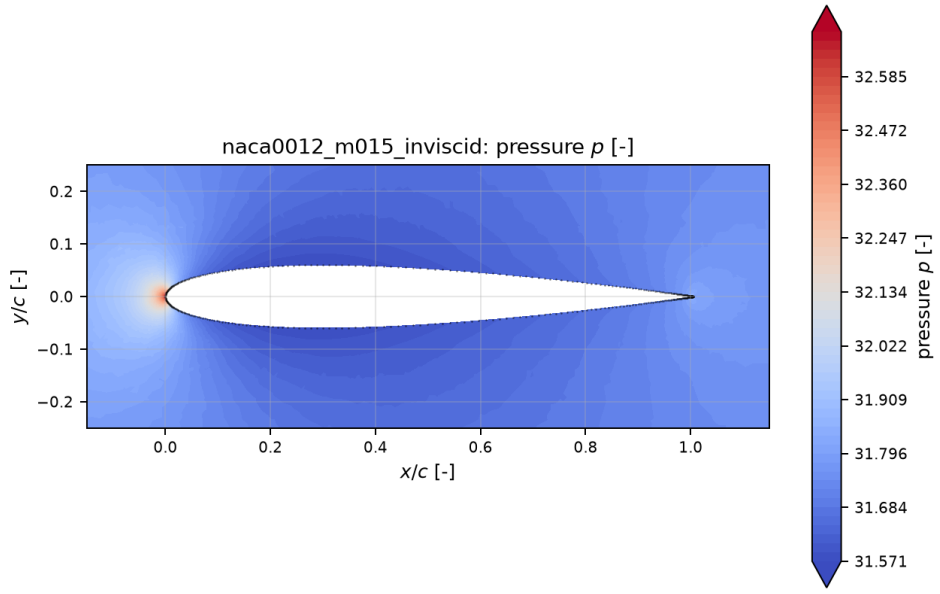


Figure 22: Pressure field for naca0012_m015_inviscid.

9.4 naca0012_m015_laminar_re5000

Submitted status: **converged**; final step 1294, final physical time 0.000, residual reduction 4.00 orders. Final forces: $C_d = 0.04643$, $C_l = -0.00472$, $C_{m,z} = -0.00106$.

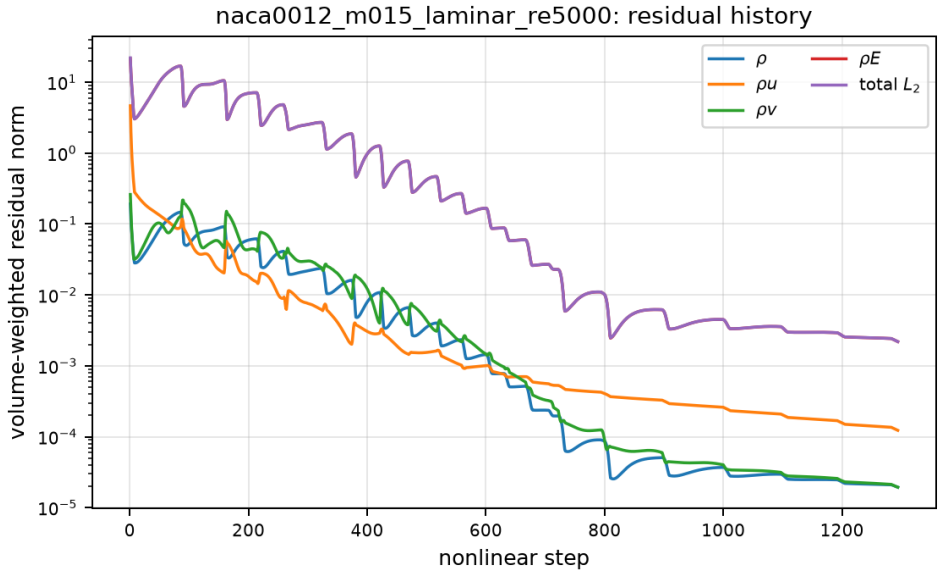


Figure 23: Residual history for naca0012_m015_laminar_re5000.

9.5 naca0012_m080_inviscid

Submitted status: **converged**; final step 14554, final physical time 0.000, residual reduction 4.00 orders. Final forces: $C_d = 0.01647$, $C_l = -0.00048$, $C_{m,z} = 0.00018$.

9.6 naca0012_m080_laminar_re5000

Submitted status: **converged**; final step 7948, final physical time 0.000, residual reduction 4.00 orders. Final forces: $C_d = 0.06853$, $C_l = -0.00245$, $C_{m,z} = -0.00049$.

9.7 naca0012_m200_inviscid

Submitted status: **converged**; final step 706, final physical time 0.000, residual reduction 3.36 orders. Final forces: $C_d = 0.09191$, $C_l = 0.00033$, $C_{m,z} = -0.00005$.

9.8 naca0012_m200_laminar_re5000

Submitted status: **converged**; final step 8177, final physical time 0.000, residual reduction 3.00 orders. Final forces: $C_d = 0.12506$, $C_l = -0.00155$, $C_{m,z} = 0.00008$.

10 Discussion and Limitations

The strongest quantitative results are the steady cylinder at $Re = 20$ (converged to more than seven orders in the global residual, with $C_d \approx 2.10$, close to the classical laminar value of about

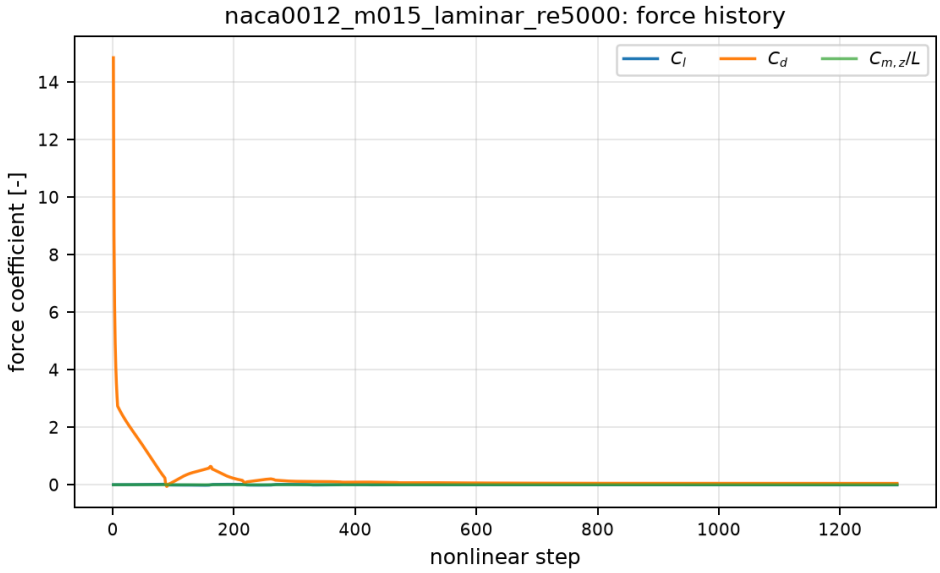


Figure 24: Force history for naca0012_m015_laminar_re5000.

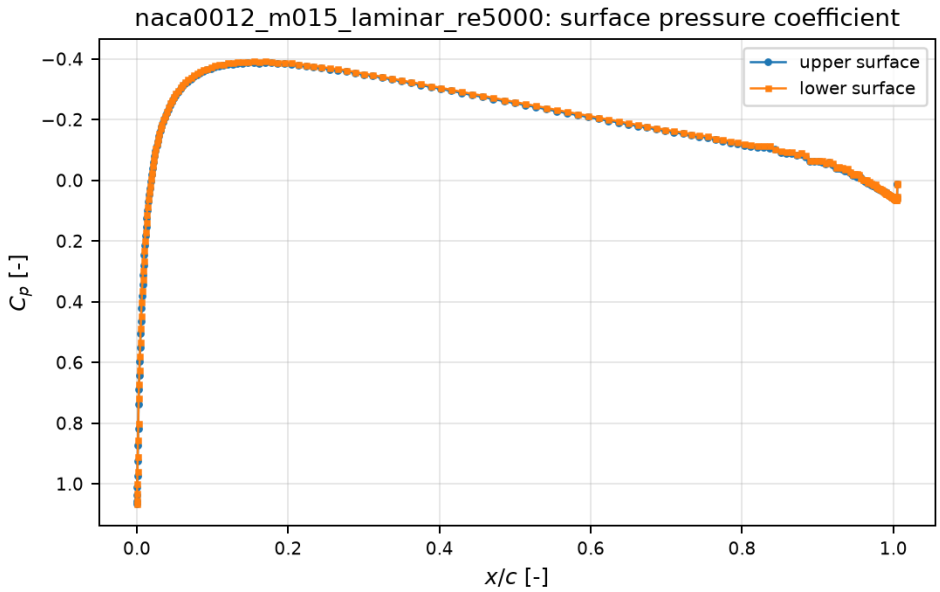


Figure 25: Surface C_p distribution for naca0012_m015_laminar_re5000.

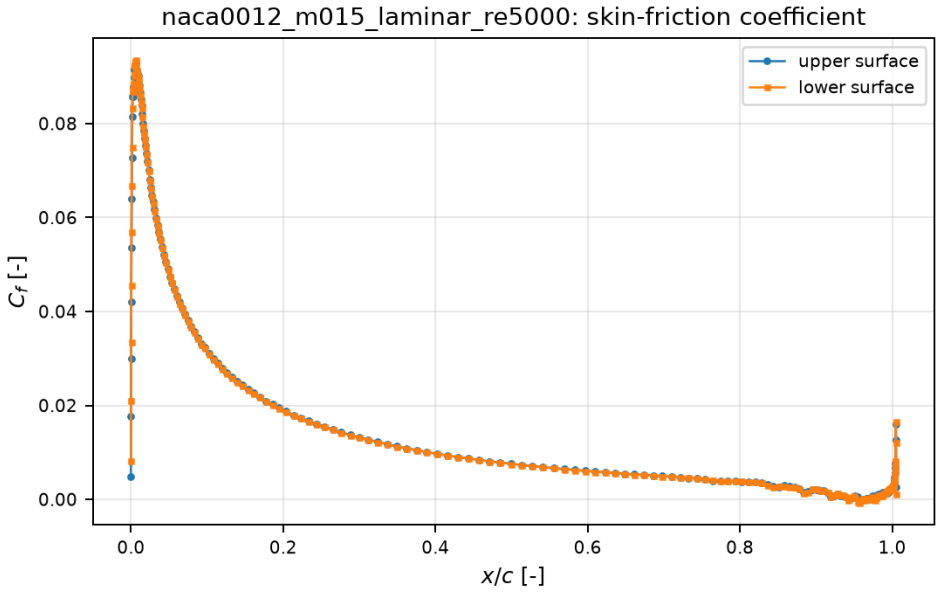


Figure 26: Surface C_f distribution for naca0012_m015_laminar_re5000.

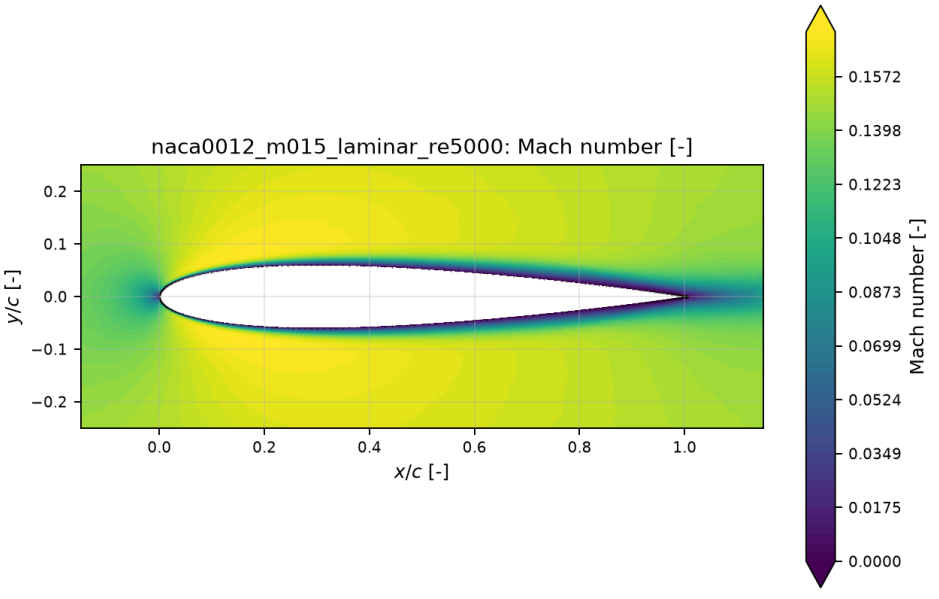


Figure 27: Mach-number field for naca0012_m015_laminar_re5000.

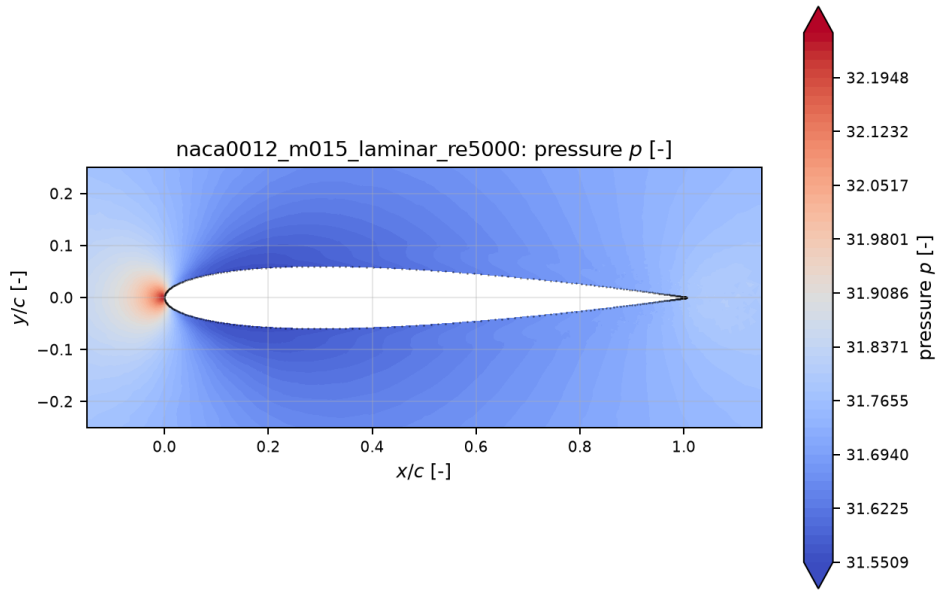


Figure 28: Pressure field for naca0012_m015_laminar_re5000.

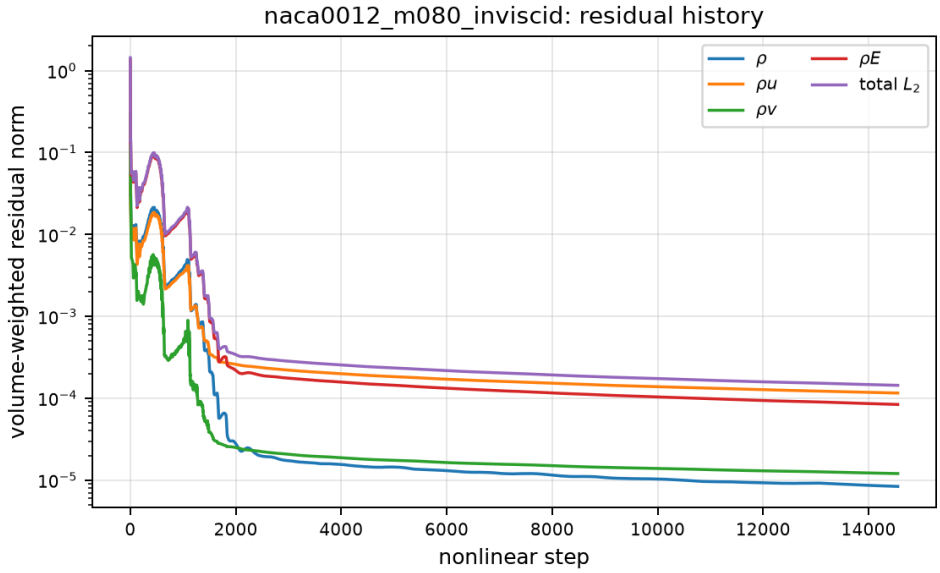


Figure 29: Residual history for naca0012_m080_inviscid.

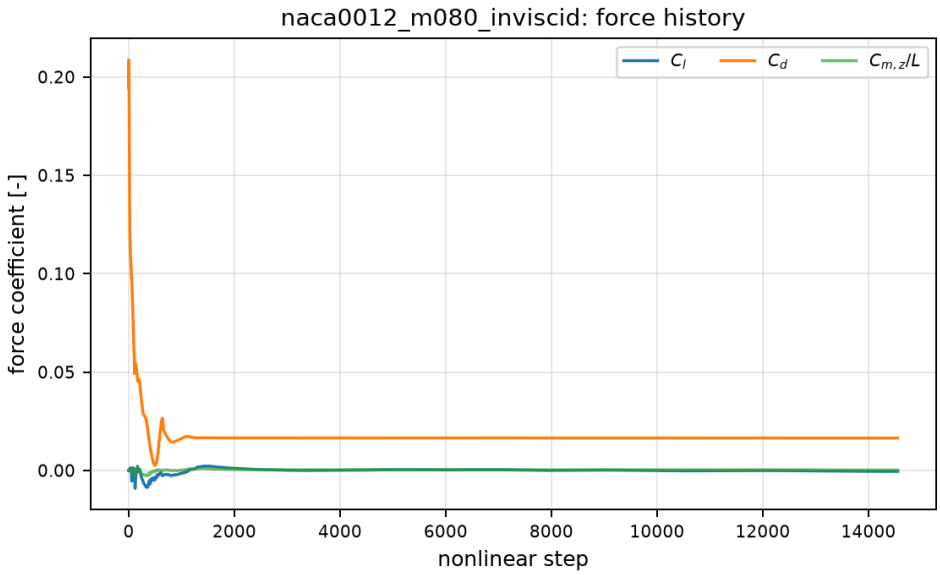


Figure 30: Force history for naca0012_m080_inviscid.

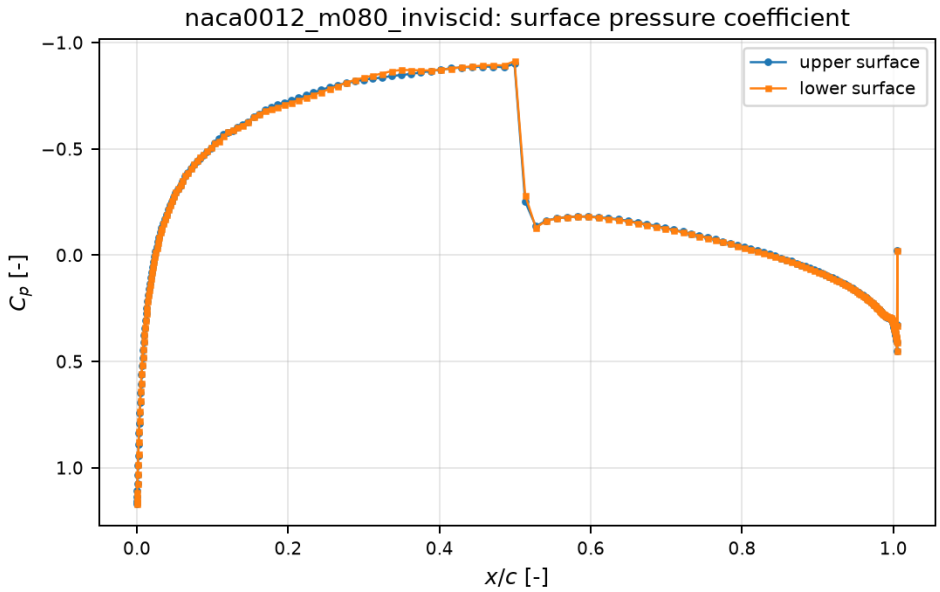


Figure 31: Surface C_p distribution for naca0012_m080_inviscid.

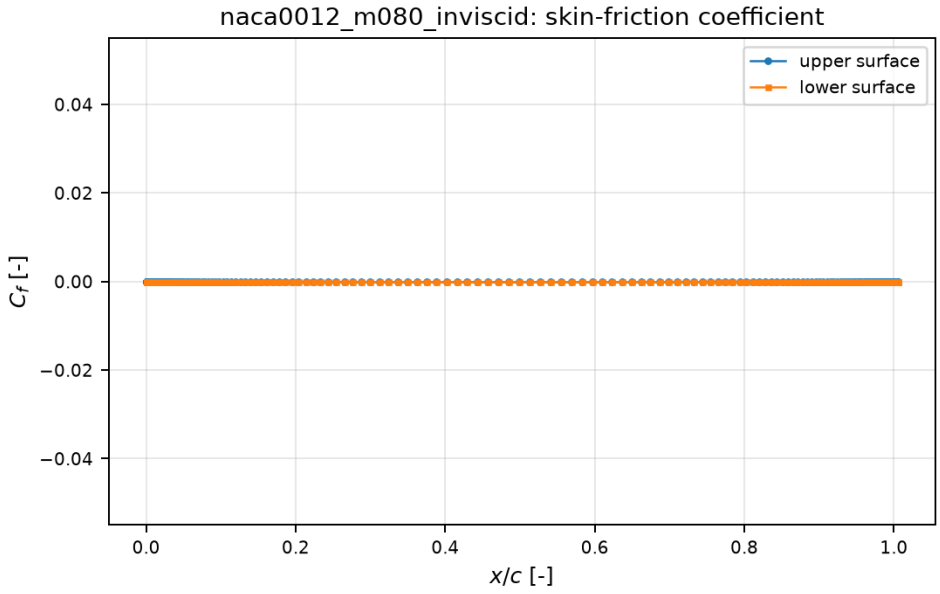


Figure 32: Surface C_f distribution for naca0012_m080_inviscid.

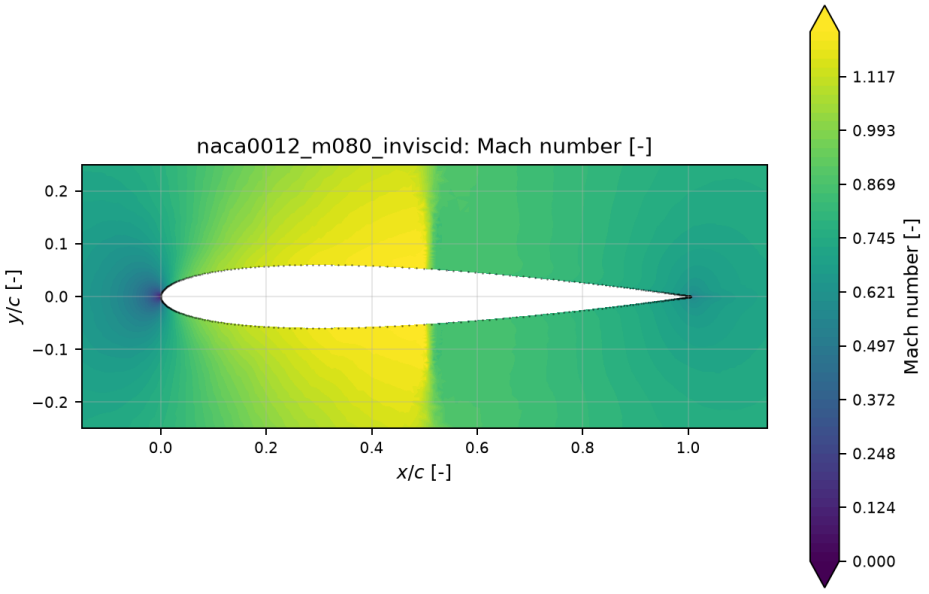


Figure 33: Mach-number field for naca0012_m080_inviscid.

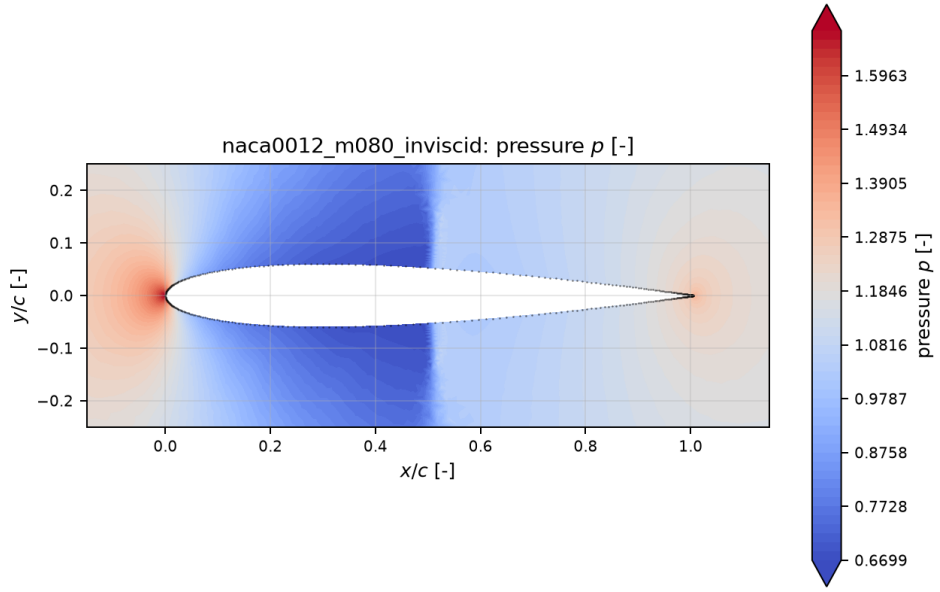


Figure 34: Pressure field for naca0012_m080_inviscid.

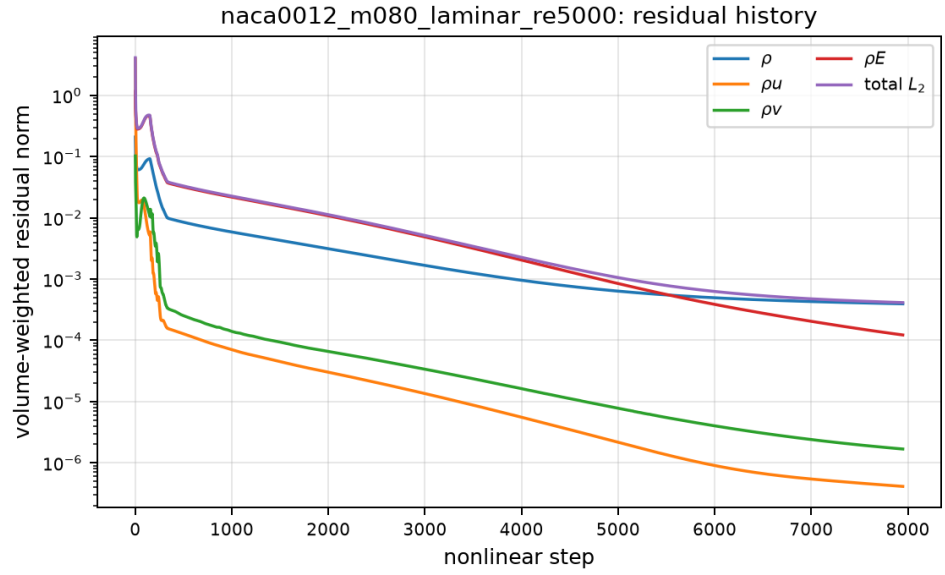


Figure 35: Residual history for naca0012_m080_laminar_re5000.

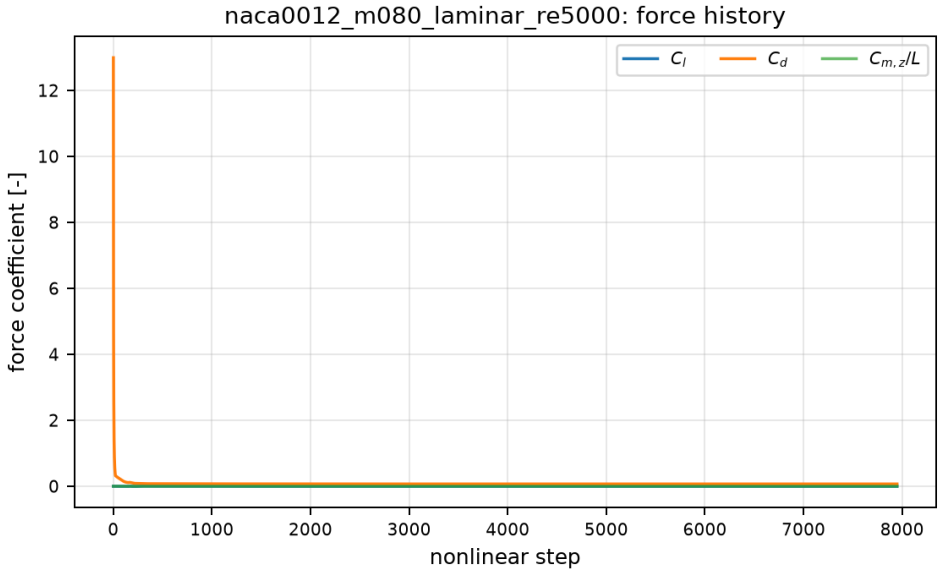


Figure 36: Force history for naca0012_m080_laminar_re5000.

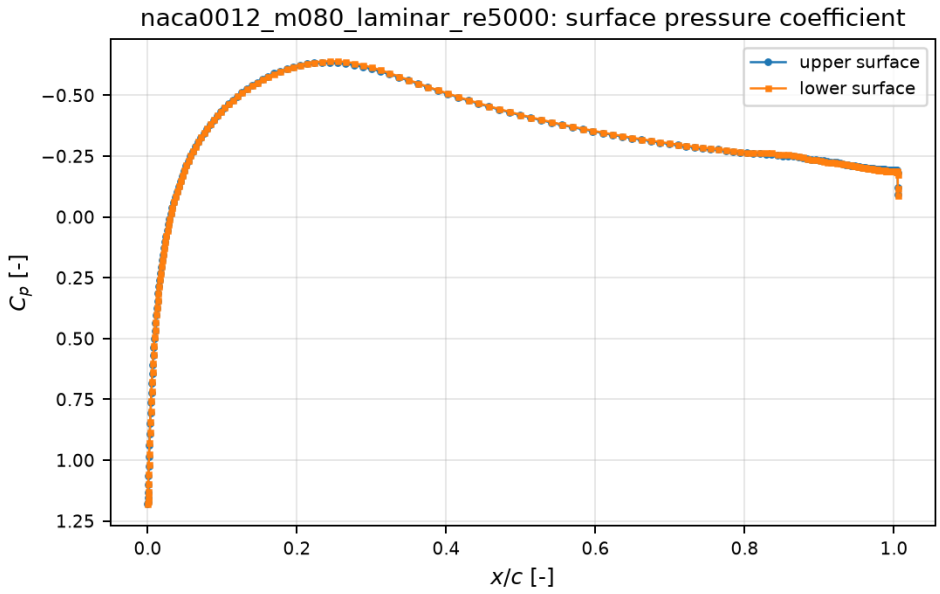


Figure 37: Surface C_p distribution for naca0012_m080_laminar_re5000.

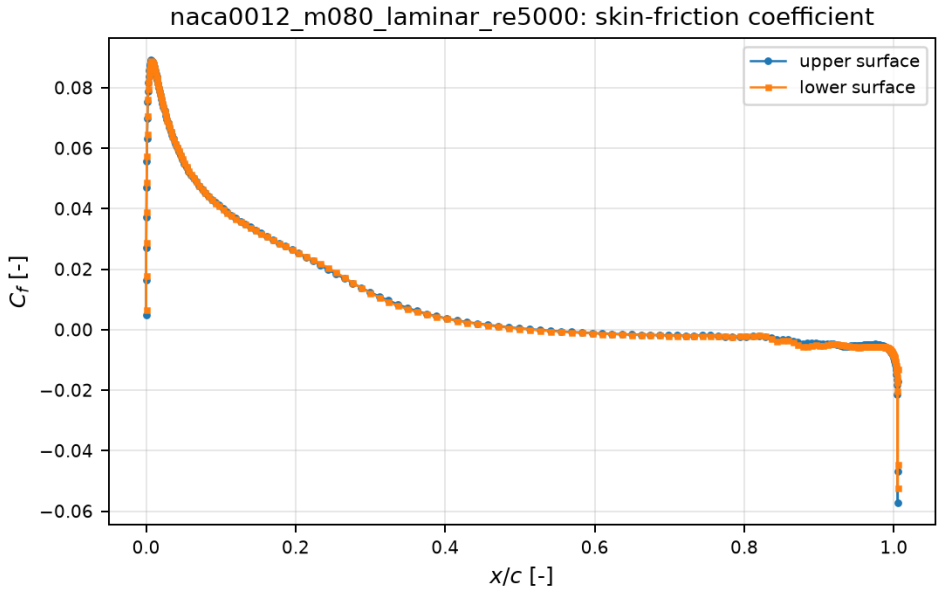


Figure 38: Surface C_f distribution for naca0012_m080_laminar_re5000.

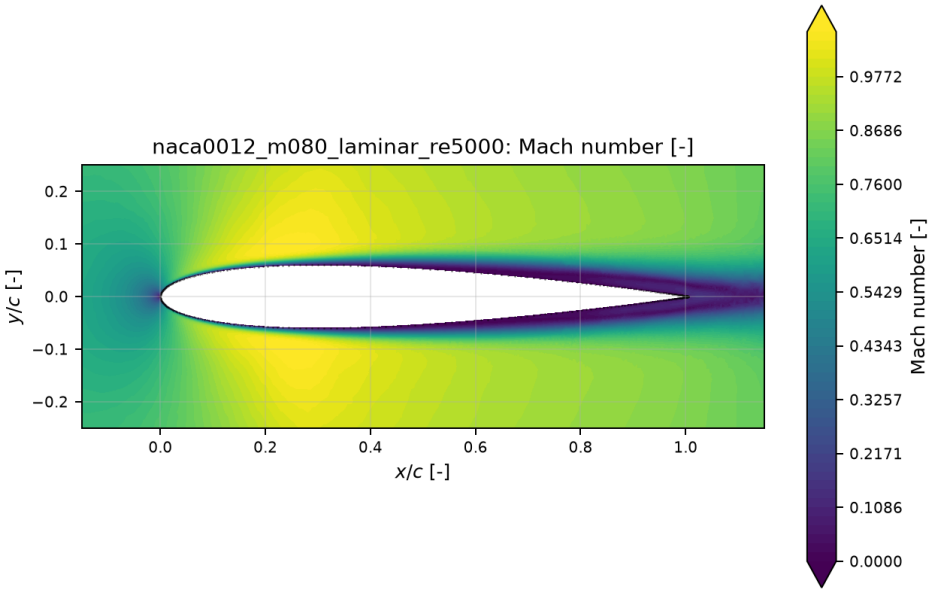


Figure 39: Mach-number field for naca0012_m080_laminar_re5000.

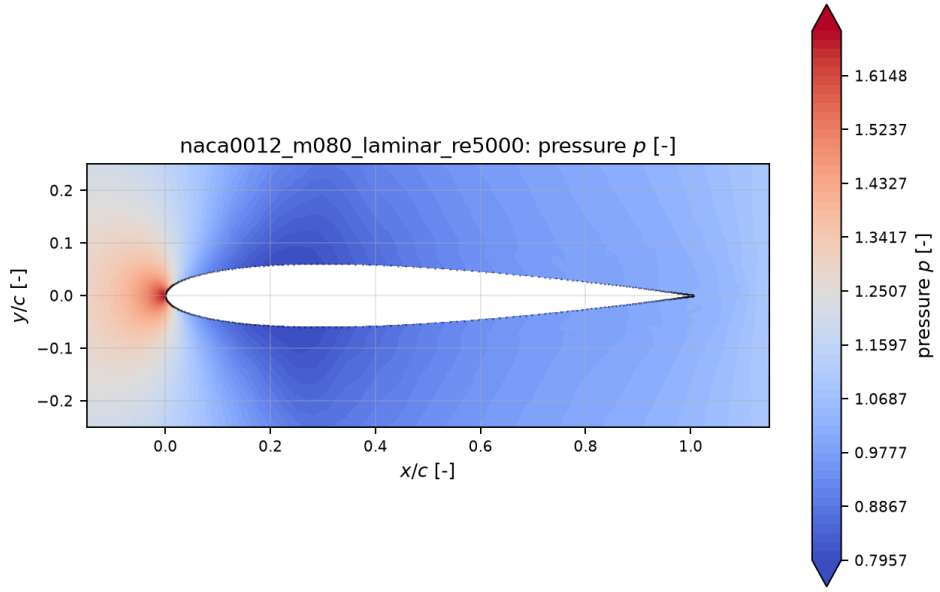


Figure 40: Pressure field for naca0012_m080_laminar_re5000.

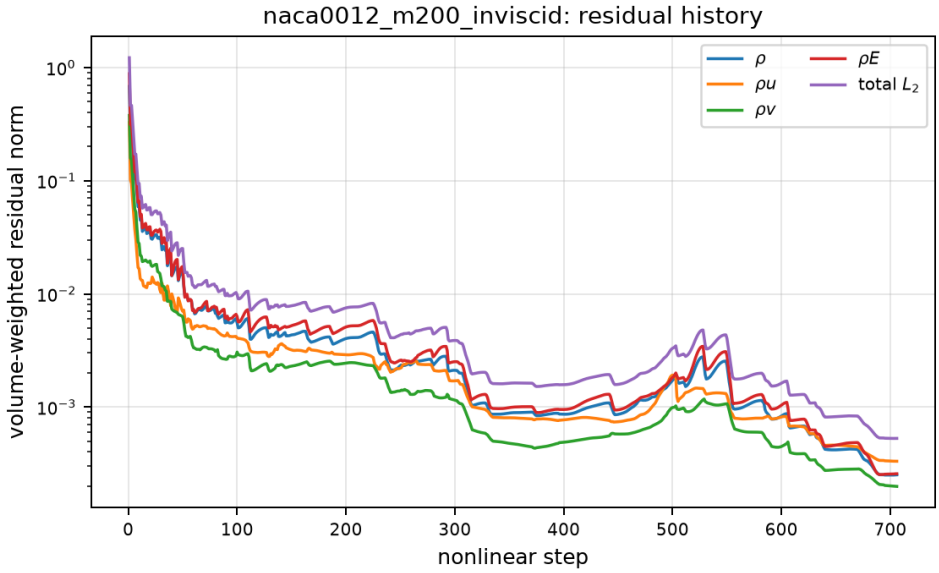


Figure 41: Residual history for naca0012_m200_inviscid.

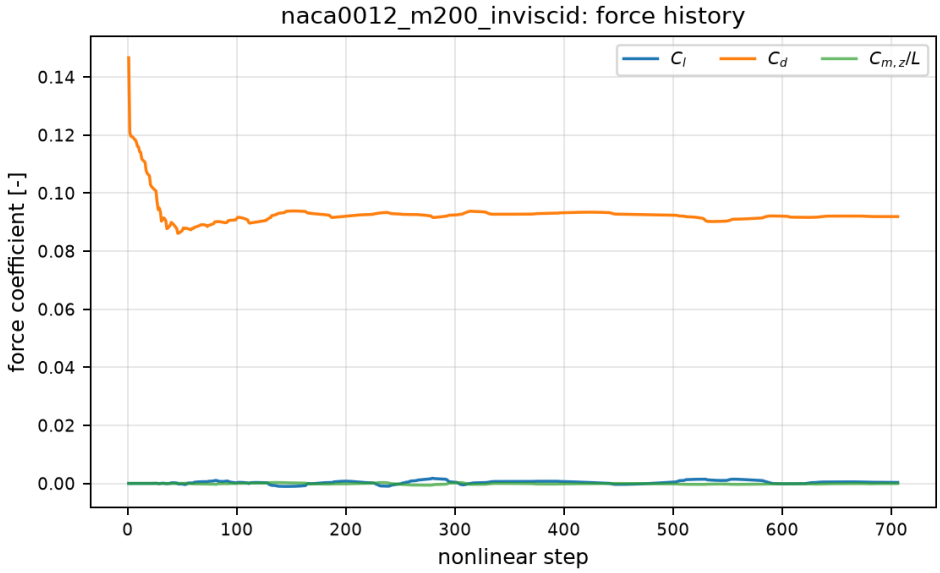


Figure 42: Force history for naca0012_m200_inviscid.

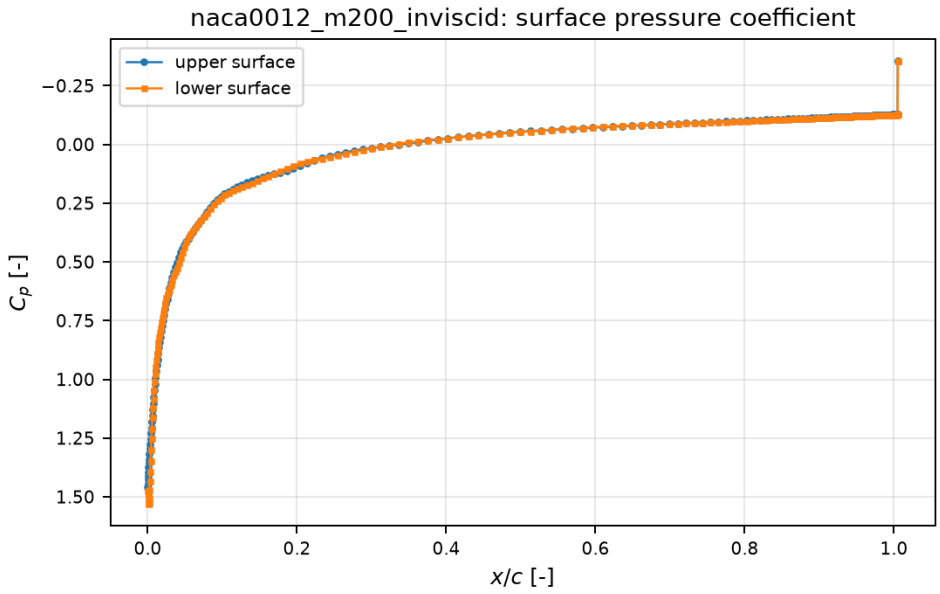


Figure 43: Surface C_p distribution for naca0012_m200_inviscid.

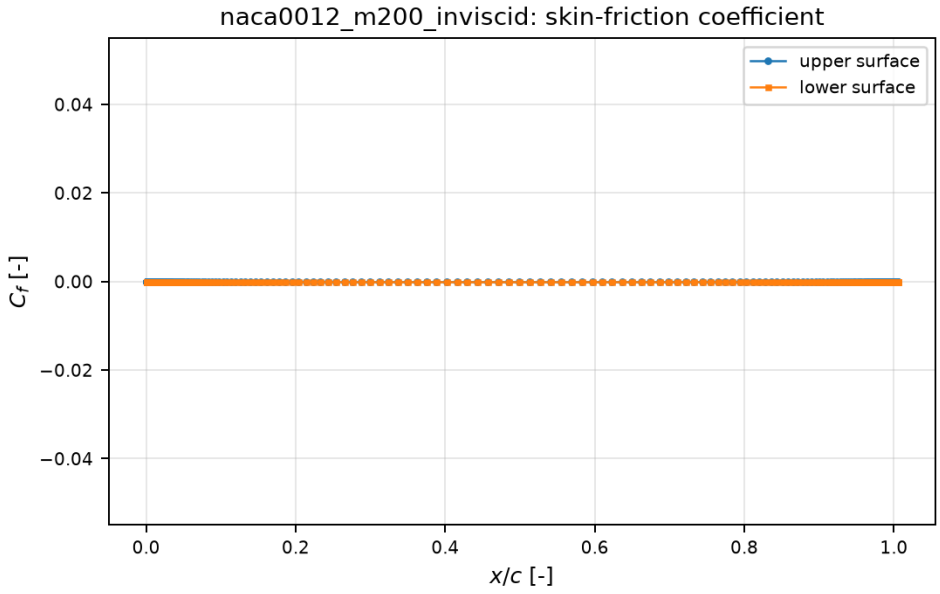


Figure 44: Surface C_f distribution for naca0012_m200_inviscid.

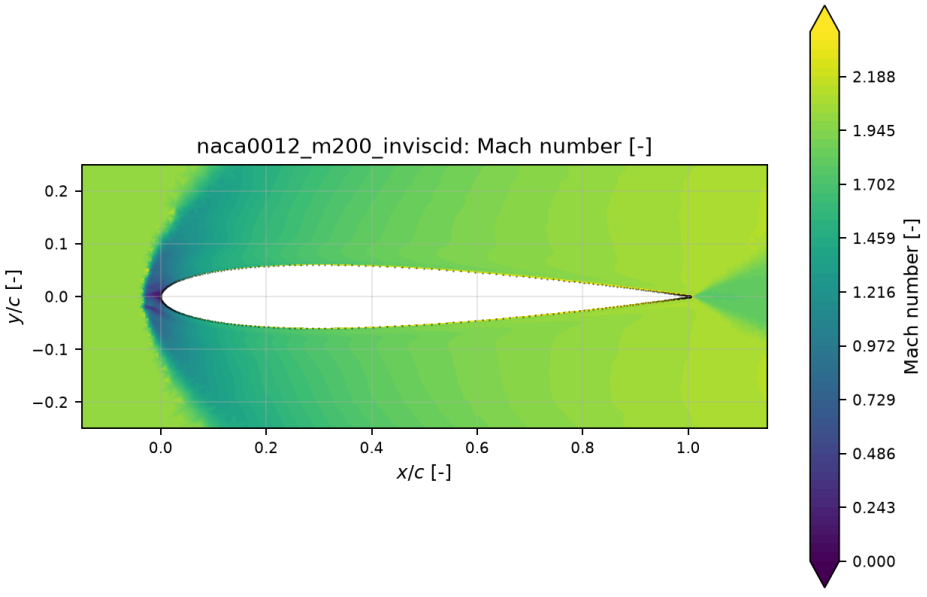


Figure 45: Mach-number field for naca0012_m200_inviscid.

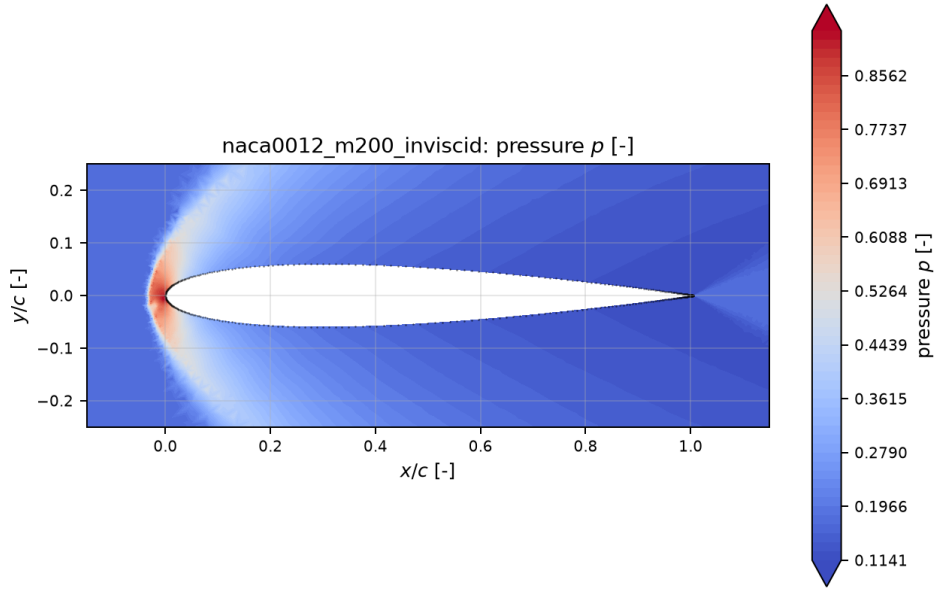


Figure 46: Pressure field for naca0012_m200_inviscid.

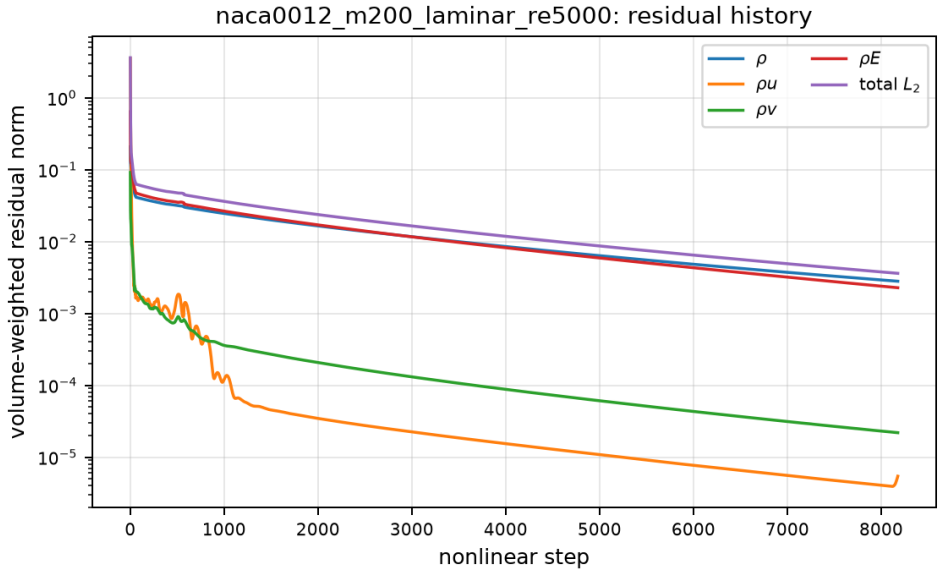


Figure 47: Residual history for naca0012_m200_laminar_re5000.

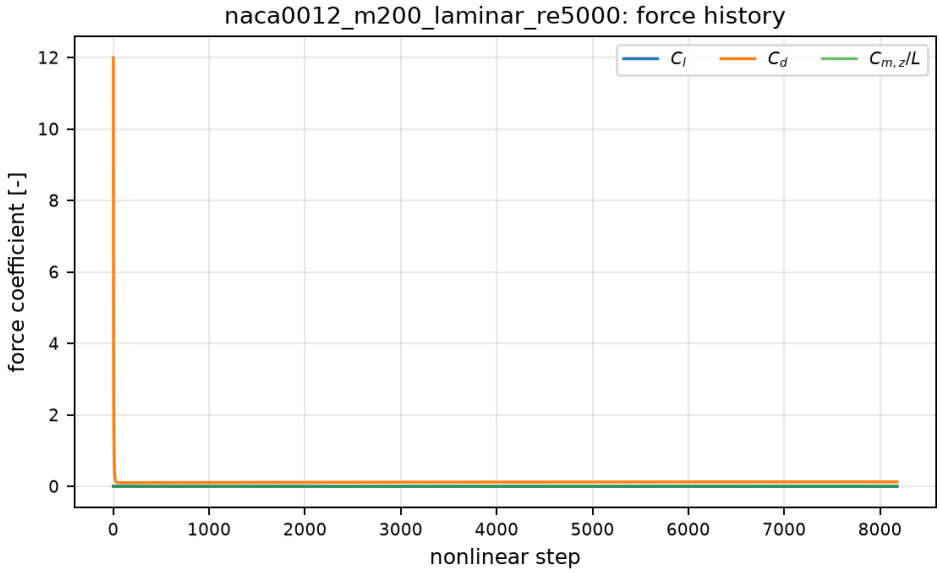


Figure 48: Force history for naca0012_m200_laminar_re5000.

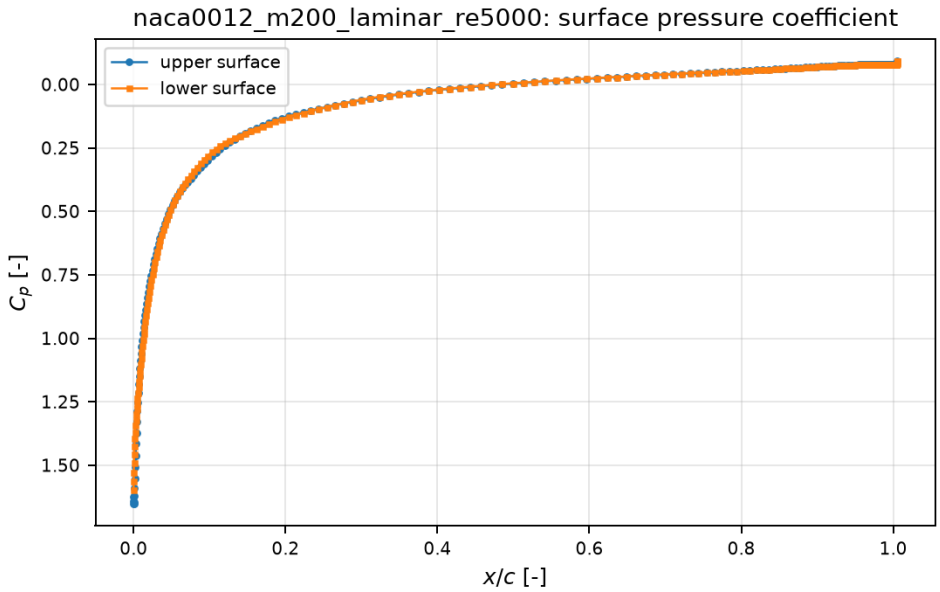


Figure 49: Surface C_p distribution for naca0012_m200_laminar_re5000.

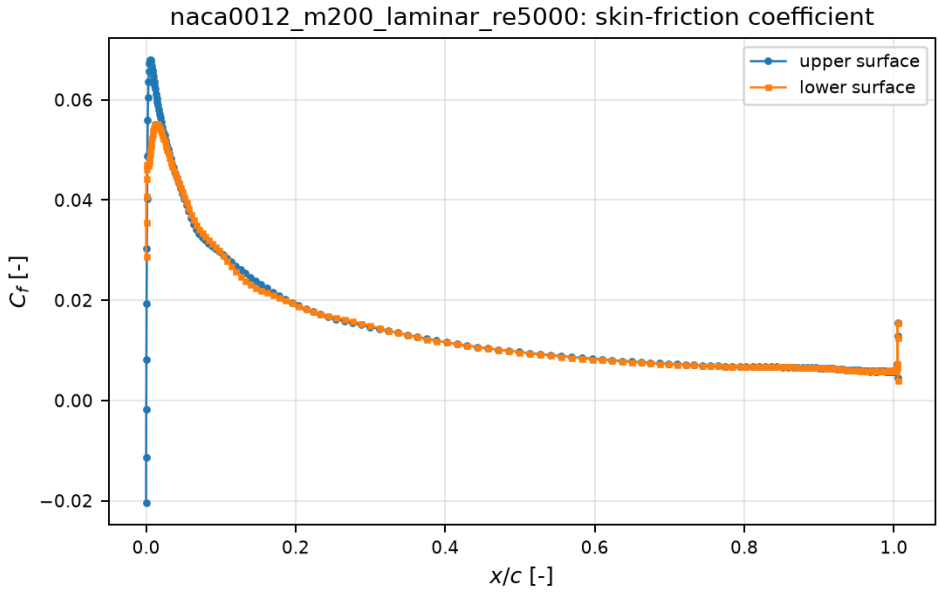


Figure 50: Surface C_f distribution for naca0012_m200_laminar_re5000.

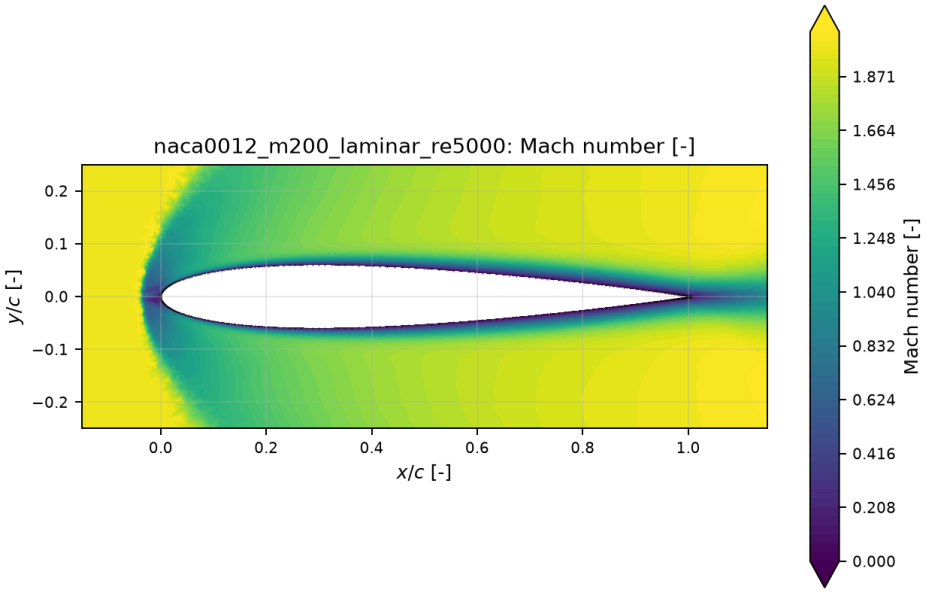


Figure 51: Mach-number field for naca0012_m200_laminar_re5000.

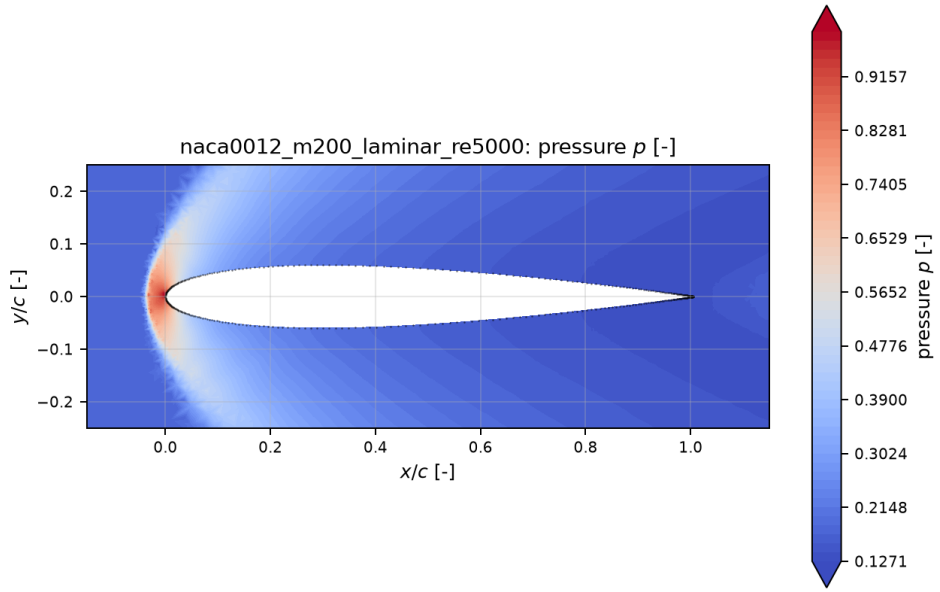


Figure 52: Pressure field for naca0012_m200_laminar_re5000.

2.0), the two transonic/supersonic inviscid NACA cases (stable shocks with the limiter active), and the transient cylinder. The Re=200 cylinder force history supports a post-transient vortex-street interpretation; the measured Strouhal number and lift amplitude are reported with the submitted force data. The main limitation is the production slope factor of 0.5. Full-slope reconstruction is implemented and is the formal second-order method, but it produced an anti-diffusive limit cycle on the highly stretched NACA boundary-layer meshes and on the cylinder. The 0.5 factor and the documented CFL caps are conservative robustness choices, not a claim of full formal second-order accuracy. The $Re = 5000$ NACA laminar drag is correspondingly higher than the finest-grid literature values; the boundary-layer mesh is extremely fine in the wall normal direction, but the reduced slope adds numerical diffusion.

One case uses an explicit first-order cell-centre fallback: the $M_\infty = 0.15$ inviscid NACA0012 case. The reduced-slope second-order reconstruction produced a nonphysical negative drag at the sharp trailing edge, where the discrete Kutta condition is sensitive; the first-order Rusanov discretization converges to a positive drag and is recorded as a fallback in that case's metadata. The first-order fallback is not used for the viscous boundary-layer cases.

The $Re = 200$ production run uses a fixed pseudo-time CFL of 5 rather than the supplied recommendation of 1. This is a documented inner-solve acceleration: the physical time step, BDF2 discretization, inner residual target, and frozen-history update are unchanged, and the actual inner convergence statistics are reported rather than hidden. The $Re = 200$ run starts from the converged $Re = 20$ field with a small asymmetry; the initial transient is discarded from the statistical analysis. The four-core cgroup quota limits the wall-clock value of the np=8 demonstration, but the halo/partition correctness is verified independently.

11 Reproducibility

The submission contains the C++ source, CMake build, preprocess/solve scripts, Python plotting and sanity-check tools, all final result packages, the LaTeX source and compiled PDF, and a run

manifest with exact commands, rank counts and wall times. The `README.md` gives the build and run sequence; `run_all_final.sh` is restartable and skips completed cases. The Python environment is a local `.venv` with numpy and matplotlib. All figures can be regenerated from the submitted CSV/VTK files with the commands in the README.

12 Conclusion

The submission provides an original, MPI-parallel unstructured finite-volume solver and a full set of computed case packages for the benchmark. The steady cases are converged or at documented plateaus with stable forces, and the Re=200 cylinder satisfies the statistical-periodicity criterion. The report states the numerical limitations and the documented deviations explicitly. No external CFD solver is wrapped or called, and no result file is synthetic.

References

[1] T. J. Barth and D. C. Jespersen, "The design and application of upwind schemes on unstructured meshes," AIAA Paper 89-0366, 1989.

[2] E. F. Toro, M. Spruce, and W. Speares, "Restoration of the contact surface in the HLL-Riemann solver," *Shock Waves*, 4:25–34, 1994.

[3] S. Yoon and A. Jameson, "Lower-upper symmetric-Gauss-Seidel method for the Euler and Navier–Stokes equations," *AIAA Journal*, 26:1025–1026, 1988.

[4] G. Karypis and V. Kumar, "A fast and high quality multilevel scheme for partitioning irregular graphs," *SIAM J. Sci. Comput.*, 20:359–392, 1998.